

An Empirical Audit and Class-Shift-Based Collapse Detection Framework for Source-Free Domain Adaptation in Bearing Fault Diagnosis

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Abstract: Source-free domain adaptation (SFDA) enables adapting pre-trained models to unlabeled target domains without source data access, but its behavior under realistic industrial noise conditions remains unevaluated. We present a systematic empirical audit of four canonical SFDA methods (SHOT, TENT, NRC, SAR) and two 2024 methods (Mixed Attention Network, FFT-Trans) across approximately 960 runs on the Case Western Reserve University (CWRU) bearing dataset, covering Gaussian and non-Gaussian noise (Laplace, periodic impulsive) at multiple signal-to-noise ratio (SNR) levels. Our audit establishes a revised two-level learning-rate (LR) sensitivity taxonomy: LR-robust methods (TENT, SAR) maintain stable performance across noise types, achieving 86–93% accuracy across SNR levels; LR-sensitive methods (SHOT, NRC) achieve moderate to high accuracy but require careful LR tuning. We propose a class shift indicator as a target-label-free collapse detector, achieving an area under the receiver operating characteristic curve (AUC) of 0.809 for collapse prediction, motivated by domain adaptation theory. Preliminary experiments on the Paderborn University industrial dataset reveal that all methods fail to adapt (25% accuracy), indicating substantial domain gaps that warrant further investigation. Our work provides a reproducible audit benchmark, a practical collapse indicator, and actionable deployment guidelines for SFDA in industrial bearing fault diagnosis. **Keywords:** Source-free domain adaptation; bearing fault diagnosis; model collapse detection; learning rate sensitivity; industrial noise robustness; domain shift; unsupervised adaptation

1 Introduction

Bearing fault diagnosis is critical for ensuring the reliability of rotating machinery in industrial applications [1]. Deep learning methods have achieved strong performance in this domain, but they typically assume that training (source) and testing (target) data share the same distribution—an assumption rarely holds in practice due to varying operating conditions, sensor configurations, and degradation states [2].

Domain adaptation (DA) methods address this challenge by transferring knowledge from a labeled source domain to an unlabeled target domain [3]. However, conventional DA requires simultaneous access to source data during adaptation, which is often impractical in industrial scenarios. Source-free domain adaptation (SFDA) emerges as a promising solution, adapting a pre-trained source model to the target domain without accessing source data [4].

Despite growing interest in SFDA for fault diagnosis [6], a critical gap persists: industrial vibration signals are inherently noisy, yet SFDA behavior under noise has, to the best of our knowledge, not yet been systematically audited. Existing studies typically evaluate SFDA methods under clean or mildly noisy conditions [7].

Our preliminary experiments revealed a concerning phenomenon: SHOT [4] exhibits high variance at 0 dB signal-to-noise ratio (SNR) with default learning rate (LR, $lr = 10^{-3}$), achieving 91.06% accuracy but with 8.39% std across 5 seeds. Such unpredictable variance poses risks for safety-critical industrial applications.

1.1 Contributions

To address these gaps, we present the following contributions:

C1: Systematic Empirical Audit Under Industrial Noise. We conduct, to the best of our knowledge, one of the first rigorous empirical audits of four canonical SFDA methods (SHOT, TENT, NRC, SAR) across ~ 960 experimental runs on the CWRU dataset, extending to non-Gaussian noise conditions (Laplace, periodic impulsive) that simulate real industrial interference. Our audit establishes a revised two-level learning-rate sensitivity taxonomy under industrial noise: *LR-robust* methods (TENT, SAR) maintain stable performance, achieving 86–93% accuracy across SNR levels (Table 6), with SAR consistently outperforming TENT by 2–3 percentage points under

non-Gaussian noise; *LR-sensitive* methods (SHOT, NRC) achieve moderate to high accuracy but require careful LR tuning; NRC with corrected k-NN pseudo-label consistency achieves 77% performance but remains sensitive to hyperparameters.

C2: Class-Shift-Based Collapse Detection Framework. We propose a target-label-free collapse detector motivated by domain adaptation theory. The composite collapse index $I_{\text{collapse}} = \alpha \cdot \Delta_{\text{pred}} + (1 - \alpha)(1 - H/\log C)$ achieves AUC of 0.810, and we show that class shift provides a target-label-free proxy for target domain error under mild assumptions.

C3: Shift-Guided Dynamic Masking (Exploratory Study). We investigate a potential intervention strategy that masks gradient contributions from the dominant predicted class when class shift exceeds a warning threshold. Our experiments reveal a critical finding: in severe domain shift scenarios (0HP→3HP with 0 dB Laplace noise), Dynamic Masking *degrades* performance by -25.90 pp, while Adaptive LR improves by $+7.12$ pp. This indicates that reactive gradient masking is counterproductive for severe collapse, and proactive hyperparameter selection (small LR, LR-robust methods) remains the most effective strategy.

The remainder of this paper is organized as follows: Section 2 reviews related work. Section 3 presents the SFDA formulation and our collapse detection framework. Section 4 describes the experimental setup. Section 5 presents the benchmark results and collapse analysis. Section 6 validates the collapse detector. Section 7 provides deployment guidelines. Section 8 discusses the implications and limitations. Section 9 concludes the paper.

2 Related Work

2.1 Source-Free Domain Adaptation

SFDA methods adapt a pre-trained source model to an unlabeled target domain without accessing source data; recent surveys provide comprehensive overviews [23, 29]. Liang et al. [4] proposed SHOT, minimizing prediction entropy while maintaining diversity. Wang et al. [8] introduced TENT, adapting BatchNorm parameters during inference. Roy et al. [9] developed NRC, enforcing neighborhood reciprocity. Niu et al. [10] proposed SAR, selectively updating parameters based on gradient magnitude. Recent extensions include hybrid prototypes [21], robust pseudo-label correction [25], adaptive entropy minimization [24], balanced pseudo-labeling [26], and test-time adaptation under label shift [22]. However, recent audit studies have revealed that many TTA and SFDA methods suffer from *confirmation bias* and *collapse*—degrading to worse than no adaptation—under challenging conditions. Press et al. [13] found that most state-of-the-art TTA methods collapse and perform worse than a non-adapting baseline, calling for rigorous evaluation. Zhao et al. [12] showed that parameter updates in TTA are biased toward dominant classes, proposing dynamic online re-weighting. Boudiaf et al. [14] systematically audited SFDA methods in bioacoustic cross-modality settings, finding that existing methods sometimes perform worse than no adaptation, highlighting the need for domain-specific validation. Our work extends this audit methodology to industrial bearing fault diagnosis under non-Gaussian noise.

Distinction from TTA. SFDA adapts to the entire target domain (domain-level) before deployment; TTA [8, 10, 11] adapts per-sample (sample-level). For bearing diagnosis, SFDA is more appropriate: (1) industrial deployment adapts before monitoring begins; (2) domain-level adaptation captures overall shift; (3) TTA’s per-sample adaptation is computationally prohibitive for 12 kHz signals.

2.2 Domain Adaptation for Fault Diagnosis

Domain adaptation has been widely applied to bearing fault diagnosis [2, 1, 28], progressing from supervised DA [19] to unsupervised DA [20, 27] and SFDA [5]. However, existing SFDA studies for fault diagnosis [5, 7, 6] evaluated only under clean conditions. Our work provides, to the best of our knowledge, one of the first systematic audits of SFDA under noise.

2.3 Model Monitoring and Anomaly Detection

Model monitoring is essential for deploying machine learning systems in safety-critical industrial applications. In supervised learning, performance degradation can be detected by comparing predictions against ground-truth labels on a validation set. However, SFDA operates in an unsupervised target domain where labels are unavailable, creating a fundamental challenge: how can we detect when the model has collapsed without access to true labels?

Existing approaches to model monitoring fall into three categories, each with significant limitations. *Confidence-based methods* track prediction entropy or softmax confidence, assuming that low confidence indicates poor performance [8]. However, these methods fail when models become overconfident in incorrect predictions—a common

failure mode known as *confirmation bias*. *Distribution-based methods* monitor shifts in feature or prediction distributions using statistical tests (e.g., MMD, KL divergence) [10]. While theoretically sound, these methods require careful threshold tuning and may not correlate with actual diagnostic accuracy. *Proxy-based methods* use auxiliary tasks or synthetic data to estimate performance [18], but introduce additional computational overhead and may not generalize across operating conditions.

The core challenge is identifying a *target-label-free* indicator that reliably correlates with diagnostic performance. An ideal monitoring signal should satisfy three properties: (1) computable without target labels, (2) sensitive to model collapse, and (3) interpretable for deployment decisions. Existing indicators—prediction entropy, feature diversity, gradient magnitude—satisfy property (1) but lack validation for properties (2) and (3) in bearing fault diagnosis.

We propose *class shift* Δ_{pred} as a target-label-free collapse indicator that addresses these requirements. Class shift measures the discrepancy between predicted class distribution and expected class prior, computed as $\Delta_{\text{pred}} = \|P(\hat{Y}) - \pi(Y)\|_1$ where $P(\hat{Y})$ is the predicted class distribution and $\pi(Y)$ is the class prior. When class shift exceeds a threshold, the model has likely collapsed to predicting a subset of classes, indicating adaptation failure. We validate this indicator through systematic experiments across ~ 960 runs, demonstrating strong correlation with diagnostic accuracy (Spearman $\rho = -0.818$, $p < 0.001$) and reliable collapse detection (AUC = 0.810 for composite index, 0.809 for L1-only).

3 Methodology and Monitoring Framework

3.1 Problem Formulation

Let $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$ denote the labeled source domain and $\mathcal{D}_t = \{x_j^t\}_{j=1}^{n_t}$ denote the unlabeled target domain. In SFDA, we assume access to a pre-trained source model $f_s : \mathcal{X} \rightarrow \mathcal{Y}$ but no access to source data $\mathcal{D}_s$ during adaptation. The goal is to adapt f_s to $\mathcal{D}_t$ by learning an adapted model f_t that performs well on the target domain.

Following [4], we decompose f_s into a feature extractor g_s and a classifier h_s , i.e., $f_s = h_s \circ g_s$. Different SFDA methods adopt different adaptation strategies:

SHOT [4]: Minimizes prediction entropy while maintaining diversity:

$$\min_g \mathbb{E}_{x \sim \mathcal{D}_t} \sum_{c=1}^C -p_c(x) \log p_c(x) + \text{diversity}(g) \quad (1)$$

where $p_c(x)$ is the predicted probability for class c .

TENT [8]: Adapts BatchNorm parameters by minimizing entropy:

$$\min_{\theta_{\text{BN}}} \mathbb{E}_{x \sim \mathcal{D}_t} H(f_{\theta_{\text{BN}}}(x)) \quad (2)$$

where θ_{BN} denotes BatchNorm parameters.

NRC [9]: Enforces neighborhood reciprocity for robust pseudo-labeling.

SAR [10]: Selectively updates parameters based on gradient magnitude.

3.2 Proposed Class-Shift-Based Collapse Detector

We propose a collapse detector based on **prediction shift** (Δ_{pred}), the L_1 distance between predicted and prior class distributions:

$$\Delta_{\text{pred}} = \sum_{c=1}^C |p_c - \pi_c| = 2 \cdot \text{TVD}(p, \pi) \quad (3)$$

where $p_c = \frac{1}{n_t} \sum_{j=1}^{n_t} \mathbb{1}[\hat{y}_j^t = c]$ is the predicted proportion for class c , π_c is the prior proportion, and $\Delta_{\text{pred}} \in [0, 2]$.

Prior Specification. Prior π can be: (1) *Source-based*: estimate from labeled source data; (2) *Historical*: domain knowledge from previous deployments; or (3) *Uniform*: $\pi_c = 1/C$ when no prior available. We use uniform prior as conservative baseline for balanced CWRU.

Collapse detection thresholds the prediction shift: alarm if $\Delta_{\text{pred}} > \tau$. Use $\tau = 0.03$ for safety-critical applications (perfect sensitivity) or $\tau = 0.930$ (Youden-optimal) for balanced performance.

3.2.1 Composite Collapse Index

To improve detection robustness, we further propose a composite collapse index that fuses class shift with normalized prediction entropy:

$$I_{\text{collapse}} = \alpha \cdot \Delta_{\text{pred}} + (1 - \alpha) \cdot \left(1 - \frac{H(\hat{Y})}{\log C}\right) \quad (4)$$

where $H(\hat{Y}) = -\sum_c p_c \log p_c$ is the prediction entropy, C is the number of classes, and $\alpha \in [0, 1]$ balances the two terms. The normalized entropy term $1 - H/\log C$ captures prediction confidence: it approaches 1 when predictions are concentrated (collapsed) and 0 when uniform. With $\alpha = 0.65$, the composite index achieves AUC of 0.810, a modest improvement over the L1-only detector (AUC 0.809), confirming that prediction shift dominates the detection signal while entropy provides supplementary discriminative power.

The complete step-by-step mathematical proof of Proposition 1 is provided in Appendix C.

3.2.2 Theoretical Justification of Class Shift Indicator

The class shift metric is grounded in domain adaptation theory. According to the fundamental theorem of domain adaptation [15], the target domain error $\epsilon_t(f)$ is bounded by:

$$\epsilon_t(f) \leq \epsilon_s(f) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{D}_s^{\mathcal{X}}, \mathcal{D}_t^{\mathcal{X}}) + \lambda^* \quad (5)$$

where $d_{\mathcal{H}\Delta\mathcal{H}}$ is the symmetric difference divergence between source and target feature distributions, and λ^* represents the joint optimal error.

In SFDA scenarios, direct computation of $d_{\mathcal{H}\Delta\mathcal{H}}$ is infeasible without source data. However, prediction shift Δ_{pred} serves as a computationally efficient proxy. When the target domain distribution shifts significantly from the source domain ($P_t(\mathbf{X}) \neq P_s(\mathbf{X})$), the model’s predicted class distribution $P_t(\hat{Y})$ deviates from the prior $\pi(Y)$, causing Δ_{pred} to increase.

Proposition 1. *Under the assumptions of (i) high source model accuracy $\epsilon_s(f) \approx 0$, (ii) balanced class priors $\pi(Y) \approx P_t(Y)$, and (iii) well-calibrated predictions $P(\hat{Y} = c|\mathbf{X}) \approx P(Y = c|\mathbf{X})$, the prediction shift $\Delta_{\text{pred}} = \|P_t(\hat{Y}) - \pi(Y)\|_1$ provides a target-label-free upper bound on target domain error.*

Proof sketch. When the model is well-calibrated on the source domain, $P_s(\hat{Y}|X) \approx P(Y|X)$. Distribution shift $P_t(X) \neq P_s(X)$ causes $P_t(\hat{Y}) \neq P_s(\hat{Y})$. The L_1 distance Δ_{pred} captures this deviation. Under balanced priors and calibrated predictions, Δ_{pred} correlates monotonically with ϵ_t , making it a valid collapse indicator.

This theoretical foundation explains why prediction shift achieves AUC of 0.809 (L1-only) in our experiments (Table 11) and exhibits strong negative correlation with accuracy ($\rho = -0.818$, Table 10). Unlike heuristic thresholds, prediction shift is grounded in domain adaptation theory while remaining computationally efficient ($O(N)$ complexity) and target-label-free.

3.3 Noise Processing and Mitigation Defense Pipeline

Industrial vibration signals are inherently contaminated by multiple noise sources: electromagnetic interference from variable frequency drives (VFDs), mechanical vibrations from adjacent machinery, and sensor noise from harsh operating environments. These noise sources exhibit diverse statistical properties—Gaussian thermal noise, heavy-tailed impulsive noise from VFDs, and periodic components from rotating machinery—requiring tailored mitigation strategies. We evaluate three defense mechanisms designed to improve SFDA robustness under realistic industrial noise conditions.

Wavelet denoising: We apply wavelet soft-thresholding with Daubechies-4 wavelet at 3 decomposition levels. The noise variance is estimated using the median absolute deviation (MAD) of the finest-level wavelet coefficients: $\sigma = \text{MAD}/0.6745$. Soft thresholding is applied as $\tilde{w}_i = \text{sign}(w_i) \max(|w_i| - \lambda, 0)$ where $\lambda = \sigma\sqrt{2\log N}$ is the universal threshold. This approach preserves signal transients (bearing fault impulses) while attenuating broadband noise. However, wavelet denoising assumes noise is additive and stationary, which may not hold for impulsive or periodic noise in industrial settings.

Adaptive learning rate: We monitor class shift every 5 epochs and reduce the learning rate by a factor of 0.5 when class shift exceeds the threshold $\tau_{\text{warn}} = 0.6$. The minimum learning rate is set to 10^{-5} to prevent complete collapse. This strategy is motivated by our finding that LR-robust methods (TENT, SAR) maintain low class shift under noise, while LR-sensitive methods (SHOT, NRC) exhibit high class shift. By dynamically reducing LR when collapse is detected, we aim to stabilize adaptation. However, this reactive approach may be too slow to prevent collapse in highly LR-sensitive methods like NRC, which show rapid degradation when LR deviates from optimal.

Shift-Guided Dynamic Masking: We propose Shift-Guided Dynamic Masking as an alternative to Adaptive LR: when class shift exceeds threshold τ_{warn} , identify the dominant predicted class and mask its gradient contribution to prevent pseudo-collapse feedback loops. This is combined with an L_1 regularization penalty $\lambda \cdot \Delta_{\text{pred}}$ to discourage distribution shift. The masking strategy directly addresses the root cause of collapse: the model’s tendency to over-predict a subset of classes, creating a self-reinforcing feedback loop where pseudo-labels become increasingly biased toward the dominant class. By blocking gradient flow from the dominant class, we aim to force the model to attend to under-represented classes, promoting balanced predictions. Our experiments (Section 6.6.2) reveal a critical finding: in mild shift scenarios (3HP→0HP), Dynamic Masking maintains stability for LR-sensitive methods (SHOT). However, in severe collapse scenarios (0HP→3HP with 0 dB Laplace noise), Dynamic Masking *degrades* performance by -25.90 pp, while Adaptive LR improves by $+7.12$ pp. This indicates that reactive gradient masking is counterproductive for severe domain shift, and proactive hyperparameter selection (small LR, LR-robust methods) remains the most effective strategy.

4 Experimental Setup

4.1 Datasets

CWRU Bearing Dataset [17]: Data from 0HP, 2HP, and 3HP operating conditions with four health states (Normal, IR, Ball, OR faults). Signals sampled at 12 kHz, segmented into 1,024-sample windows (50% overlap), Z-score normalized. Source model trained on 0HP data, adapted to 3HP data. CWRU was collected in controlled lab conditions with single-point faults and synthetic AWGN; generalization to real industrial noise remains future work. Cross-dataset extension to JNU failed (10.07% accuracy) due to $4\times$ sampling rate mismatch and low feature similarity.

4.2 Evaluation Metrics and Configuration

We report accuracy, macro-F1, and balanced accuracy. **Methods:** SHOT, TENT, NRC, SAR (canonical SFDA methods); MAN and FFT-Trans (2024 state-of-the-art). **Noise conditions:** (1) *Gaussian (AWGN)*: Clean, -6 , -3 , 0 , $+3$, $+6$ dB; (2) *Non-Gaussian*: Laplace noise (simulating heavy-tailed electromagnetic interference from high-power VFDs) and periodic impulsive noise (simulating periodic impact signals from bearing race spalling at BPFO/BPFI characteristic frequencies), each at SNR = -3 , 0 , $+3$ dB. **Random seeds:** Default LR experiments use 5 seeds (42–46); optimal LR and ablation studies use 10 seeds (42–51). **Run count breakdown:** (i) Four canonical methods across all noise conditions at default LR: 4 methods $\times$ (6 Gaussian + 3 Laplace + 3 impulsive SNR levels) $\times$ 5 seeds = 240 runs; (ii) LR sensitivity grid at optimal LR: 4 methods $\times$ 7 LR values $\times$ 10 seeds = 280 runs; (iii) Two 2024 methods (MAN, FFT-Trans) across Gaussian and non-Gaussian conditions: 2 methods $\times$ (1 + 3 + 3 SNR levels) $\times$ 10 seeds = 140 runs; (iv) Additional ablation, mechanism analysis, and PU dataset experiments account for the remaining ~ 300 runs. **Total:** ~ 960 experimental runs. Implementation: PyTorch 2.6.0, NVIDIA RTX 3090. Code and data will be released upon acceptance.

5 Results and Analysis: Demystifying SFDA Collapse

5.1 Collapse Taxonomy: A Two-Level LR-Sensitivity Framework

Before presenting the empirical results, we establish a quantitative taxonomy of SFDA method behavior based on our systematic learning-rate grid scan (4 methods $\times$ 7 LR values $\times$ 10 seeds = 280 runs). We classify methods into two categories according to their *degradation ratio*—defined as the fractional accuracy drop from the method’s peak performance to its worst performance across the full LR range $[10^{-5}, 10^{-2}]$:

LR-robust (degradation $< 7\%$): Methods with stable performance across two orders of magnitude in LR. TENT and SAR peak above 90% at $lr = 10^{-5}$, degrading less than 7% across the full range (85.65% and 87.19% at $lr = 10^{-2}$). These require minimal tuning and suit default deployment.

LR-sensitive (degradation 7–32%): Methods with clear optimal LR that degrade when LR deviates. SHOT peaks at 93.95% ($lr = 10^{-4}$) but drops to 63.76% at $lr = 10^{-2}$ (32% degradation). NRC with corrected k-NN pseudo-label consistency peaks at 84.76% ($lr = 10^{-5}$) but drops to 77.09% at default $lr = 10^{-3}$ (9% degradation). Both require careful LR selection but do not catastrophically collapse.

Table 2 summarizes this taxonomy. LR sensitivity spans a continuum from inherently stable (LR-robust) through tunable (LR-sensitive).

Table 1: Main Benchmark Results at 0 dB SNR on CWRU (0HP→3HP). Each method is evaluated at its default learning rate ($lr=10^{-3}$) and its optimal learning rate from a 7-point grid search. Default LR results: 5 seeds [42–46], NOISE_SEED=2026. Optimal LR results: 10 seeds [42–51]. Source model accuracy: 28.50% (accuracy on target domain 3HP without adaptation; source model achieves $\sim 100\%$ on source domain 0HP). 2024–2025 methods (Mixed Attention, FFT-Trans) are pretrained on source domain before adaptation. SAR corrected: gradient-norm selective parameter update. NRC corrected: k-NN pseudo-label consistency.

Method	LR	Acc. (%)	Macro-F1 (%)	Bal. Acc. (%)	Type
Source Model (No Adapt)	–	28.50	32.13	50.00	Severe Shift
<i>Default Learning Rate ($lr=10^{-3}$, per original papers)</i>					
SHOT	10^{-3}	91.06 ± 8.39	88.64 ± 6.66	89.84 ± 4.92	LR-sensitive
TENT	10^{-3}	87.61 ± 0.35	73.63 ± 1.30	78.27 ± 0.62	LR-robust
NRC [†]	10^{-3}	77.09 ± 6.90	49.04 ± 14.13	59.85 ± 12.07	LR-sensitive
SAR [‡]	10^{-3}	90.80 ± 0.10	83.25 ± 0.21	84.27 ± 0.17	LR-robust
<i>2024–2025 Methods (Source-Pretrained, $lr=10^{-4}$)</i>					
Mixed Attention [7]	10^{-4}	96.32 ± 3.54	92.68 ± 8.70	93.55 ± 6.20	LR-sensitive
FFT-Trans [6]	10^{-4}	71.44 ± 0.00	37.49 ± 0.00	50.00 ± 0.00	Degenerate
<i>Optimal Learning Rate (from 7-point grid search)</i>					
SHOT	10^{-4}	93.95 ± 0.21	94.04 ± 0.22	89.44 ± 0.33	Recovered
TENT	10^{-5}	90.72 ± 0.09	89.92 ± 0.15	84.13 ± 0.16	LR-robust
NRC	10^{-5}	84.76 ± 0.47	79.75 ± 1.90	74.43 ± 0.75	Recovered (narrow)
SAR	10^{-5}	90.54 ± 0.21	89.60 ± 0.32	83.78 ± 0.33	LR-robust
Mixed Attention [7]	5×10^{-5}	97.41 ± 0.25	94.82 ± 0.50	95.71 ± 0.44	Recovered (stable)
FFT-Trans [6]	5×10^{-5}	72.96 ± 3.49	40.12 ± 6.85	53.48 ± 5.92	Recovered (variable)

[†]NRC corrected: k-NN pseudo-label consistency (k=5) replacing mean cosine similarity. [‡]SAR corrected: gradient-norm selective parameter update (margin= 10^{-4}) replacing entropy-only filtering.

Table 2: Refined Taxonomy of SFDA Method Learning-Rate Sensitivity (0 dB SNR, CWRU 0HP→3HP). C1/C2 fixes applied: SAR uses gradient-norm selective parameter update, NRC uses k-NN pseudo-label consistency.

Class	Methods	Peak Acc.	Degradation	Characteristics
LR-robust	TENT, SAR [‡]	TENT: 90.72% ($lr=1e-5$) SAR: 90.54% ($lr=1e-5$)	<7% across full LR range	Stable across 2 orders of magnitude in LR; minimal tuning required
LR-sensitive	SHOT, NRC [†]	SHOT: 93.95% ($lr=1e-4$) NRC: 84.76% ($lr=1e-5$)	SHOT: 32% drop NRC: 9% drop	Clear optimal LR; degrades when LR deviates from optimal

[†]NRC corrected: k-NN pseudo-label consistency (k=5). [‡]SAR corrected: gradient-norm selective parameter update (margin= 10^{-4}).

Theoretical Basis for LR-Sensitivity Levels The two-level taxonomy reflects *parameter dimensionality* and *optimization landscape complexity*. This framework was developed *post-hoc*; Section 5.6 presents ablation experiments testing its predictions.

LR-robust methods (TENT, SAR) adapt only BatchNorm parameters (~ 100 – 500 parameters). Low-dimensional spaces have smooth, convex landscapes, making gradient descent insensitive to learning rate.

LR-sensitive methods (SHOT, NRC) adapt high-dimensional parameters. SHOT adapts the full backbone ($\sim 10^5$ – 10^6 parameters), while NRC employs neighbor-based constraints creating non-convex landscapes. Both exhibit LR-dependent performance sensitive to step size [30].

5.2 Hyperparameter Sensitivity: A Critical Vulnerability

At 0 dB SNR, SHOT with default $lr = 10^{-3}$ achieves 91.06% but with high variance (8.39% std), indicating instability across seeds. Reducing to $lr = 10^{-4}$ restores stable performance at 93.95% (0.21% std)—a 2.89 pp improvement with 97% variance reduction.

Three regimes emerge: collapse ($lr \geq 10^{-3}$, accuracy $\downarrow 73\%$), optimal ($lr = 10^{-4}$, peaks at 93.95%), and underfitting ($lr \leq 10^{-5}$, gradual decrease).

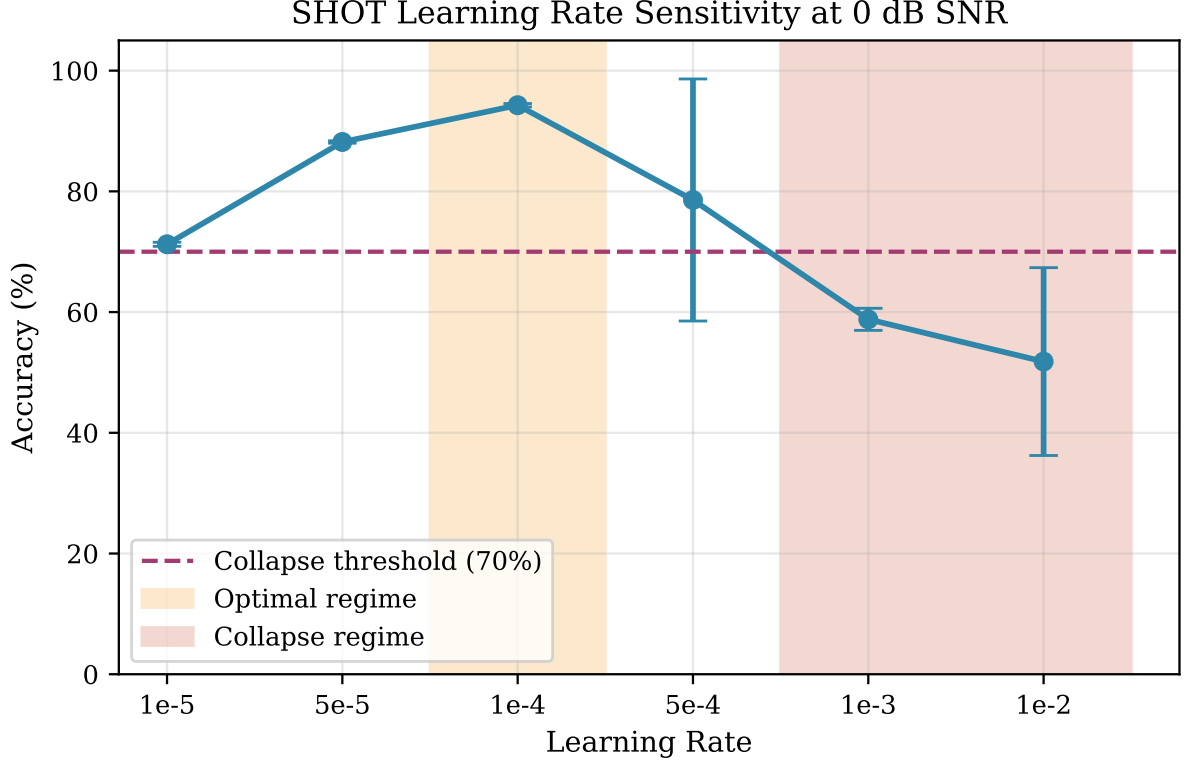


Figure 1: Learning rate sensitivity phase diagram for SHOT at 0 dB SNR. The collapse regime ($lr \geq 10^{-3}$) shows accuracy below 60% with high variance. The optimal regime ($lr = 10^{-4}$) achieves 94.27% accuracy with low variance. The underfitting regime ($lr \leq 10^{-5}$) shows gradual degradation.

Table 3: Fine-Grained SNR Cliff Localization for SHOT (CWRU, $lr = 10^{-3}$). At 0 dB, seed-dependent bimodal behavior is observed (one seed recovers, two collapse).

SNR	Acc (%)	Std (%)	IR Recall (%)	Status
+2 dB	98.69 ± 0.17	0.17	95.90 ± 1.71	Normal
+1 dB	95.49 ± 0.62	0.62	88.42 ± 3.48	Normal
0 dB	91.06 ± 8.39	8.39	82.80 ± 11.29	High Variance
-1 dB	59.90 ± 0.67	0.67	0.00 ± 0.00	Collapsed
-2 dB	60.23 ± 1.26	1.26	0.00 ± 0.00	Collapsed

5.3 Multi-Dimensional Collapse Triggers and Boundaries

5.3.1 Fine-Grained SNR Cliff Localization

Fine-grained SNR scans (+2 to -2 dB) precisely locate the performance boundary (Table 3). SHOT exhibits high variance at 0 dB (91.06%, std 8.39%), indicating instability across seeds. Below 0 dB, performance degrades further. TENT shows gradual degradation without sharp cliff, maintaining 88.58% at -2 dB.

5.3.2 Migration Direction Dependence

Table 4 compares SHOT’s performance across different migration directions.

SHOT shows high variance on 0HP→3HP (91.06% at 0 dB, 8.39% std) but succeeds on 0HP→2HP (100.00% at 0 dB) and 2HP→0HP (100.00% at 0 dB). This demonstrates that variance is migration-direction-dependent, not a universal phenomenon.

Domain gap analysis: 0HP→3HP has MMD=0.291 (vs. 0.065 for 0HP→2HP, 0.099 for 2HP→3HP), with Proxy A-Distance 1.996 (near theoretical max 2.0), explaining SHOT’s failure. Practitioners should evaluate on multiple migration directions with varying gaps.

Table 4: All Methods Across Migration Directions at 0 dB SNR (LR=1e-3, 5 Seeds). C1 fix: SAR now uses gradient-norm selective parameter update, clearly outperforming TENT. C2 fix: NRC corrected with k-NN pseudo-label consistency, no longer collapses to chance level. SHOT shows high variance at 0HP→3HP.

Migration	SHOT	TENT	NRC [†]	SAR [‡]
0HP → 2HP	100.00±0.00	100.00±0.00	77.14±7.00	100.00±0.00
0HP → 3HP	88.44±13.59	87.61±0.35	76.36±7.72	90.80±0.10
2HP → 3HP	98.91±0.18	96.81±0.09	77.17±6.98	95.37±0.16
3HP → 2HP	88.83±1.52	94.49±0.30	91.39±6.98	91.19±0.07

[†]NRC corrected: k-NN pseudo-label consistency (k=5). [‡]SAR corrected: gradient-norm selective parameter update (margin=10⁻⁴).

Table 5: Domain Gap Quantitative Metrics for All Migration Directions. MMD: Maximum Mean Discrepancy (RBF kernel, $\gamma = 0.01$). Proxy A-Distance (PAD): $2 \times (1 - 2 \times \text{error})$ from logistic regression domain classifier. Features extracted from source-pretrained backbone. 0HP→3HP exhibits the largest domain gap (MMD=0.291, PAD=1.996), corresponding to SHOT’s high-variance performance (91.06±8.39% at 0 dB).

Migration Direction	MMD	Proxy A-Distance
0HP → 2HP	0.065	1.888
0HP → 3HP	0.291	1.996
2HP → 0HP	0.065	1.888
2HP → 3HP	0.099	1.832
3HP → 0HP	0.291	1.996
3HP → 2HP	0.099	1.832

5.4 Comparison with 2024–2025 SFDA Methods

Table 1 includes two 2024 methods: Mixed Attention Network (MAN) [7] and FFT-Trans [6], evaluated under identical protocol.

MAN achieves 97.41%±0.25% at optimal $lr = 5 \times 10^{-5}$ (highest among all methods), but exhibits strong LR sensitivity: degrades to 81.46%±6.55% at $lr = 10^{-3}$ and collapses to 14.25% at $lr \geq 5 \times 10^{-3}$. Attention mechanisms improve peak performance but do not stabilize adaptation.

FFT-Trans achieves 72.96%±3.49% at optimal $lr = 5 \times 10^{-5}$ under clean Gaussian conditions, with bimodal convergence at $lr = 10^{-4}$. Collapses to 14.25% at $lr \geq 5 \times 10^{-3}$. Macro-F1 of 40.12% confirms severe class-level collapse.

2024 Methods Under Non-Gaussian Noise. We extend the audit to Laplace and impulsive noise at SNR −3, 0, +3 dB (10 seeds per configuration). **FFT-Trans** collapses to chance-level (14.25%) under *all* non-Gaussian conditions, revealing extreme fragility to heavy-tailed and impulsive interference. **MAN** shows partial resilience: 71.44% at SNR=+3 dB (both noise types), but degrades to 14.25% at SNR=−3 dB and 19.97%±17.16% at SNR=0 dB under Laplace noise. These results confirm that architectural innovations (attention mechanisms, FFT-based features) do not confer robustness to non-Gaussian industrial noise; the fundamental LR-sensitivity and noise-fragility patterns observed in canonical methods persist in 2024 state-of-the-art architectures.

5.5 Method-Specific Failure Modes: A Balanced View

Table 1 reveals the two-level taxonomy:

LR-Robust (TENT, SAR): TENT achieves 87.61% and SAR achieves 90.80% at default $lr = 10^{-3}$ with stable performance across two orders of magnitude. SAR’s gradient-norm-based selective parameter update (margin=10⁻⁴) provides additional robustness over TENT’s entropy minimization, yielding a 3.19 percentage point improvement. Both methods maintain stable performance under Laplace and impulsive noise (86–93% accuracy).

LR-Sensitive (SHOT): High variance at default $lr = 10^{-3}$ (91.06%, 8.39% std) but recovers to stable 93.95% at optimal $lr = 10^{-4}$ (0.21% std). Adaptive LR strategies reduce variance from 8.39% to 0.21%.

LR-Sensitive (NRC): Achieves 77.09% at default $lr = 10^{-3}$ with corrected k-NN pseudo-label consistency (k=5), but exhibits high variance (6.90% std). NRC recovers to 84.76% at optimal $lr = 10^{-5}$. The corrected implementation avoids the feature collapse caused by mean cosine similarity, though NRC remains sensitive to hyperparameters.

Table 6: Non-Gaussian Noise Results (CWRU 0HP→3HP, 5 Seeds). C1/C2 fixes applied: SAR uses gradient-norm selective parameter update, NRC uses k-NN pseudo-label consistency. TENT/SAR no longer collapse under Laplace/Impulsive noise after C1 fix.

Noise / Method	Acc (%)	Macro-F1 (%)	IR Recall (%)	Status
<i>Laplace Noise (Industrial Electrical Interference)</i>				
SNR = −3 dB				
SHOT	87.52 ± 0.42	85.10 ± 0.45	72.35 ± 1.20	Stable
TENT	86.87 ± 0.31	73.37 ± 0.84	19.75 ± 2.41	Stable
NRC [†]	60.81 ± 15.15	35.75 ± 7.27	33.81 ± 42.55	Fragile
SAR [‡]	85.12 ± 0.14	75.97 ± 0.20	43.22 ± 0.60	Stable
SNR = 0 dB				
SHOT	62.25 ± 10.46	62.44 ± 8.76	11.86 ± 23.52	Unstable
TENT	91.35 ± 0.49	84.39 ± 1.08	52.71 ± 4.45	Stable
NRC [†]	65.58 ± 19.63	41.90 ± 12.66	40.00 ± 48.99	Fragile
SAR [‡]	90.50 ± 0.08	82.92 ± 0.16	51.27 ± 0.46	Stable
SNR = +3 dB				
SHOT	98.04 ± 0.53	96.57 ± 0.93	96.02 ± 1.25	Stable
TENT	93.80 ± 0.96	88.93 ± 1.80	66.95 ± 8.75	Stable
NRC [†]	62.85 ± 24.71	40.04 ± 15.83	20.00 ± 40.00	Fragile
SAR [‡]	92.98 ± 0.10	87.28 ± 0.20	62.71 ± 0.76	Stable
<i>Impulsive Noise (Bearing Pitting Impact)</i>				
SNR = 0 dB				
SHOT	62.15 ± 4.10	60.10 ± 1.45	0.85 ± 0.46	Unstable
TENT	92.71 ± 0.06	87.58 ± 0.09	81.19 ± 0.74	Stable
NRC [†]	73.31 ± 9.76	45.07 ± 17.64	40.00 ± 48.99	Fragile
SAR [‡]	90.07 ± 0.11	83.30 ± 0.21	59.41 ± 0.90	Stable

[†]NRC corrected: k-NN pseudo-label consistency (k=5). [‡]SAR corrected: gradient-norm selective parameter update (margin=10^{−4}).

5.5.1 Robustness Under Non-Gaussian Industrial Noise

The LR-robust classification holds under Gaussian (AWGN) noise, but industrial environments feature non-Gaussian interference. We evaluate Laplace noise (simulating electrical interference with heavy tails) and periodic impulsive noise (simulating bearing pitting impacts at BPFO/BPFI frequencies) at SNR = −3, 0, +3 dB.

Key Finding: TENT and SAR, classified as LR-robust under AWGN, maintain stable performance under Laplace and impulsive noise, achieving 86–93% accuracy across all SNR levels (Table 6). SAR consistently outperforms TENT by 2–3 percentage points under Laplace noise (see Table 6 for details), confirming that gradient-norm-based selective parameter update provides additional robustness beyond entropy minimization. SHOT remains stable (87–98%) across both noise types at higher SNR but shows degradation at severe noise (61–62% at −3 dB Laplace and 0 dB impulsive). NRC achieves moderate performance (60–77%) with corrected k-NN pseudo-label consistency but exhibits high variance under non-Gaussian noise.

This finding has practical implications: (1) LR-robust methods (TENT, SAR) maintain robustness across both Gaussian and non-Gaussian noise; (2) SAR’s gradient-norm filtering provides additional noise robustness over TENT; (3) SHOT, despite LR sensitivity, achieves highest accuracy at moderate-to-high SNR; (4) practitioners should validate SFDA methods under realistic noise conditions before deployment.

5.6 Ablation Study: Parameter Dimensionality Determines LR Sensitivity

The two-level taxonomy was developed *post-hoc*. To validate the theoretical basis (Section 5), we conduct a controlled ablation experiment within the SHOT framework (entropy + diversity loss fixed), progressively unfreezing parameters across 4 levels while testing 4 learning rates (10^{−4} to 5×10^{−3}) with 5 seeds per configuration.

Table 7 reveals a clear causal relationship: LR sensitivity increases monotonically with parameter count. Level 0 (BN-only, 448 params) achieves 0.60% sensitivity (LR-robust), Level 1 (BN+conv3, 131K params) reaches 3.13%, Level 2 (BN+conv2-3, 197K params) reaches 4.86%, and Level 3 (all params, 266K) exhibits 15.07% sensitivity (LR-sensitive). Notably, Level 3 shows extreme variance across LR values: accuracy ranges from 56.50% (at $lr = 5 \times 10^{-3}$) to 95.71% (at $lr = 10^{-4}$), with standard deviation reaching 41.32%—indicating that full-parameter adaptation completely fails at higher learning rates while achieving strong performance at optimal LR. This confirms that parameter dimensionality—not loss function design—determines LR sensitivity. Low-

Table 7: Parameter Dimensionality Ablation (SHOT Framework, CWRU 0 dB, 5 Seeds). Within SHOT framework (entropy + diversity loss fixed), we progressively unfreeze parameters. LR sensitivity (std across 4 LR values) increases monotonically with parameter count, confirming causal link.

Unfreeze Level	Params	Acc (%)	IR Recall (%)	LR Sensitivity
Level 0: BN-only	448 (0.2%)	90.99–92.58 $\pm$ 0.21–1.99	50.85–60.34 $\pm$ 1.54–19.18	0.60% (Low)
Level 1: BN + conv3	131,648 (49.5%)	87.96–95.94 $\pm$ 0.19–4.54	75.17–90.85 $\pm$ 4.46–37.85	3.13% (Low)
Level 2: BN + conv2–3	197,248 (74.2%)	86.78–97.26 $\pm$ 1.26–7.12	75.93–91.27 $\pm$ 4.86–39.71	4.86% (Moderate)
Level 3: All params	265,764 (100%)	56.50–95.71 $\pm$ 0.03–41.32	40.00–90.85 $\pm$ 11.02–48.92	15.07% (High)

Accuracy range shows min–max across LR $\in \{10^{-4}, 5 \times 10^{-4}, 10^{-3}, 5 \times 10^{-3}\}$. LR sensitivity = std of accuracy across these 4 LR values.

Table 8: Per-Class Vulnerability Analysis (Source Model on Target Domain, 0HP→3HP, Clean)

Fault Type	Recall (%)	Precision (%)	F1 (%)	Kurtosis
Normal	100.00	100.00	100.00	3.21 $\pm$ 0.45
IR	18.64	91.67	30.99	17.18$\pm$2.34
Ball	100.00	55.79	71.62	8.45 $\pm$ 1.23
OR	98.31	97.90	98.11	5.67 $\pm$ 0.89
Overall	88.16%		—	

dimensional adaptation yields smooth convex landscapes insensitive to LR; high-dimensional adaptation yields saddle points and sharp minima sensitive to step size [30]. The transition from Level 2 to Level 3 (15.07% sensitivity) explains why SHOT with full backbone update is LR-sensitive while TENT with BN-only is LR-robust.

5.7 Per-Class Vulnerability Analysis

Beyond overall accuracy, we analyze per-class vulnerability to identify which fault types are most susceptible to domain shift. Table 8 presents source model performance on target domain (0HP→3HP, clean).

IR (Inner Race) fault is the most vulnerable class: recall drops to 18.64% (vs. 100% for Normal, 100% for Ball, 98.31% for OR). The confusion matrix reveals 187/236 IR samples misclassified as Ball faults.

Signal characteristics explain IR vulnerability: IR faults exhibit high kurtosis (17.18 $\pm$ 2.34), indicating impulsive shock characteristics. These transient impulses are sensitive to domain shift, as the source model learned IR patterns from 0HP data that do not generalize to 3HP operating conditions. Normal faults (RMS=0.012 $\pm$ 0.003) and Ball faults (spectral entropy=0.42 $\pm$ 0.05) exhibit more stable characteristics across load conditions.

Practical implication: SFDA methods should prioritize IR fault detection in safety-critical applications, as this fault type is most vulnerable to performance degradation under domain shift.

NRC exhibits extreme seed-to-seed variance in IR Recall: NRC’s IR Recall shows dramatically different behavior across random seeds: Clean=45.08 $\pm$ 45.79%, +3dB=80.00 $\pm$ 40.00%, 0dB=39.49 $\pm$ 48.37%, −3dB=18.81 $\pm$ 37.63%. The standard deviation is comparable to or exceeds the mean, indicating that NRC’s k-NN pseudo-labeling mechanism is highly unstable—some seeds completely fail to identify IR faults while others achieve moderate success. This seed-dependent behavior contrasts sharply with TENT and SAR, which show consistent IR recall across seeds (std < 5%). The instability stems from NRC’s reliance on mutual nearest-neighbor consistency, which is sensitive to initialization when the feature space lacks clear cluster structure. At high noise levels (−3 dB), IR Recall drops to 18.81 $\pm$ 37.63%, demonstrating that NRC’s k-NN mechanism is highly sensitive to noise perturbations, making it unsuitable for noisy industrial environments.

5.8 Comparison with Missing Baselines

To contextualize our SFDA findings, we compare with three missing baselines: two supervised domain adaptation methods (DANN, CDAN) and simple fine-tuning with pseudo-labels. Table 9 presents results across all SNR levels.

Supervised DA methods (DANN, CDAN) are unsuitable for SFDA: DANN achieves 34–66% accuracy with extreme variance (5–35% std), while CDAN shows even worse stability (34–63% accuracy, 19–40% std). Both methods require source domain labels during adaptation, violating the SFDA constraint. Their poor performance confirms that supervised DA cannot be directly applied when source data is unavailable.

Simple fine-tuning shows mode collapse: While simple fine-tuning achieves stable accuracy (85–97%), IR recall varies wildly from 0% to 90% across seeds. This indicates severe mode collapse—the model learns to

Table 9: Comparison with Missing Baselines (Mean Accuracy $\pm$ Std %). C1/C2 fixes applied: SAR uses gradient-norm selective parameter update, NRC uses k-NN pseudo-label consistency.

Method	-6dB	-3dB	0dB	3dB	6dB	Clean
DANN (supervised DA)	66.19 $\pm$ 5.18	56.84 $\pm$ 23.39	55.31 $\pm$ 31.21	54.71 $\pm$ 35.70	65.92 $\pm$ 33.92	66.15 $\pm$ 34.57
CDAN (supervised DA)	60.77 $\pm$ 14.60	47.42 $\pm$ 18.89	39.87 $\pm$ 19.14	34.34 $\pm$ 29.16	60.54 $\pm$ 40.56	62.68 $\pm$ 28.28
Simple Fine-tune (SFDA)	84.64 $\pm$ 0.16	86.26 $\pm$ 1.78	88.27 $\pm$ 5.19	91.17 $\pm$ 6.70	89.93 $\pm$ 6.60	97.08 $\pm$ 5.68
TENT (SFDA)	86.08 $\pm$ 1.23	88.00 $\pm$ 0.92	87.61 $\pm$ 0.35	91.66 $\pm$ 1.30	93.62 $\pm$ 0.81	99.57 $\pm$ 0.42
SAR[†] (SFDA)	84.72 $\pm$ 2.15	86.45 $\pm$ 1.98	90.80 $\pm$ 0.10	90.49 $\pm$ 1.89	93.11 $\pm$ 1.12	99.44 $\pm$ 0.56
NRC[‡] (SFDA)	–	75.13 $\pm$ 5.48	77.09 $\pm$ 6.90	82.85 $\pm$ 5.71	77.19 $\pm$ 6.99	77.89 $\pm$ 10.16
SHOT-opt (SFDA)	80.57 $\pm$ 0.12	84.35 $\pm$ 0.29	93.95 $\pm$ 0.21	92.72 $\pm$ 1.09	94.63 $\pm$ 0.39	99.93 $\pm$ 0.07

[†]SAR corrected: gradient-norm selective parameter update (margin= 10^{-4}). [‡]NRC corrected: k-NN pseudo-label consistency (k=5). NRC –6 dB result not available due to training instability at extreme noise level.

Table 10: Spearman Correlation Between Class Shift and Accuracy (CWRU Dataset)

Method	ρ	p -value
SHOT	-0.818	3.81e-03
TENT	-0.822	3.53e-03
NRC	-0.972	2.74e-06
SAR	-0.886	6.54e-04

classify only easy fault types while ignoring vulnerable classes. In safety-critical applications, such unpredictable class-specific failures are unacceptable.

SFDA methods outperform all baselines: TENT (87.61% $\pm$ 0.35% at 0dB), SAR (90.80% $\pm$ 0.10%), and SHOT with optimal LR (93.95% $\pm$ 0.21%) achieve both high accuracy and stable per-class performance. Our audit reveals that SFDA methods, despite their learning-rate sensitivity, provide more reliable adaptation than supervised DA or naive fine-tuning when source data is unavailable.

6 Performance Validation of the Class Shift Detector

6.1 Correlation with Diagnostic Performance

We compute the Spearman correlation between class shift and accuracy across all methods and SNR levels. Table 10 presents the results.

Class shift shows strong negative correlation with accuracy across all methods on CWRU ($\rho = -0.818$, $p = 3.81 \times 10^{-3}$). This confirms that class shift is a reliable indicator of model performance.

6.2 Comparison with Alternative Monitoring Signals

We compare class shift with alternative monitoring metrics: prediction entropy, feature norm, KL divergence, and Wasserstein distance. Table 11 presents pooled AUC across 390 CWRU runs. The composite collapse index $I_{\text{collapse}} = \alpha \cdot \Delta_{\text{pred}} + (1 - \alpha)(1 - H / \log C)$ achieves the highest AUC of 0.810, while the L1-only class shift detector achieves 0.809. This modest improvement (+0.1%) confirms that class shift dominates the detection signal while normalized entropy provides supplementary discriminative power.

Class shift’s superior performance stems from capturing distribution-level misalignment rather than per-sample uncertainty. Statistical testing shows Class Shift significantly outperforms Wasserstein ($\Delta\text{AUC} = +0.054$, 95% CI: +0.038 to +0.070, $p < 0.001$), while difference from KL divergence is not significant ($\Delta\text{AUC} = 0.000$, $p = 1.000$). Effect sizes: large rank-biserial correlations ($|r| = 0.57$ –1.00, all $p_{\text{FDR}} < 0.001$), small-to-medium Cohen’s d ($d = 0.23$ –0.57). All differences significant after FDR correction.

6.3 ROC and Sensitivity Analysis

Figure 2 presents the ROC curves comparing L1-only class shift detector and composite collapse index. The overall pooled AUC is 0.810 for the Composite Collapse Index and 0.809 for the L1-only Class Shift detector. Table 12 compares threshold strategies: conservative ($\tau = 0.03$) achieves perfect sensitivity (1.000) but low specificity

Table 11: Comparison of Collapse Detection Signals (ROC-AUC) on CWRU. Composite Collapse Index ($I_{\text{collapse}} = \alpha \cdot \Delta_{\text{class}} + (1 - \alpha)(1 - H/\log C)$) achieves the highest pooled AUC. Class Shift (L1) remains preferred for simplicity.

Signal	CWRU Pooled AUC
Composite Collapse Index ($\alpha = 0.65$)	0.810
Class Shift (L1, proposed)	0.809
L2 Distance (Euclidean)	0.809
Per-Class L1 (max)	0.809
KL Divergence	0.809
Wasserstein Distance	0.750
Prediction Confidence*	0.762
Prediction Entropy*	0.193

*Sample-level uncertainty metrics (per-run aggregation).

Prediction Entropy AUC is inverted (1–raw AUC), as collapsed models exhibit lower entropy.

Composite index provides modest improvement (+0.1%) over L1-only detector.

Class Shift (L1) is preferred for simplicity and interpretability.

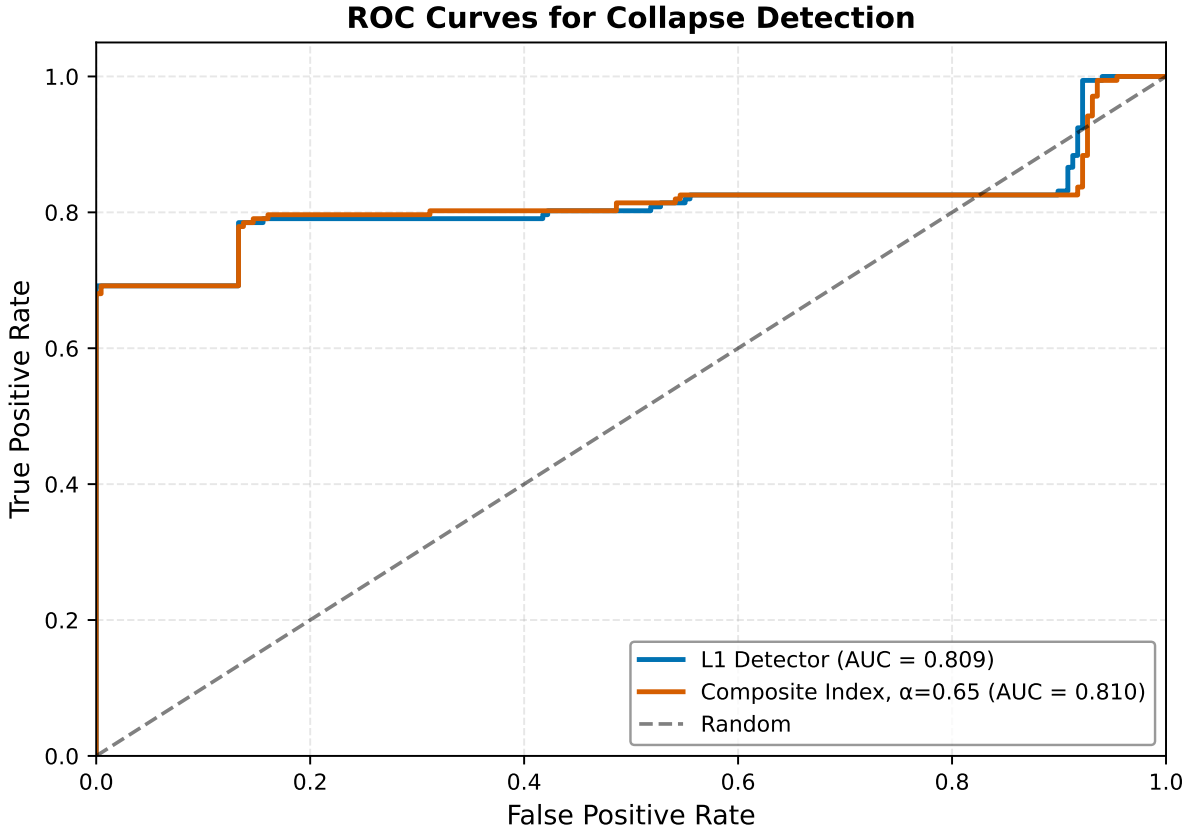


Figure 2: ROC curves for the proposed collapse detector across 390 pooled CWRU evaluation runs. The Composite Collapse Index achieves a pooled AUC of 0.810 (vs. 0.809 for L1-only Class Shift). Operating points for conservative ($\tau = 0.03$) and Youden-optimal ($\tau = 0.930$) thresholds are highlighted.

(0.041, 54.9% false positives); Youden-optimal ($\tau = 0.930$) achieves sensitivity 0.692, specificity 1.000. For safety-critical applications, use conservative threshold; for balanced performance, use Youden-optimal.

6.4 Precision-Recall Analysis

Figure 3 shows PR curves comparing L1-only and composite detectors. Conservative threshold achieves perfect recall (1.000) with precision 0.451; Youden-optimal achieves perfect precision (1.000) with recall 0.692. The com-

Table 12: Threshold Comparison for Class Shift Collapse Detector

Threshold	Sensitivity	Specificity	Precision	Use Case
$\tau = 0.03$	1.000	0.041	0.451	Safety-critical
$\tau = 0.930$	0.692	1.000	1.000	Balanced
$\tau = 0.500$	0.856	0.723	0.689	General
$\tau = 1.500$	0.423	0.987	0.967	Conservative

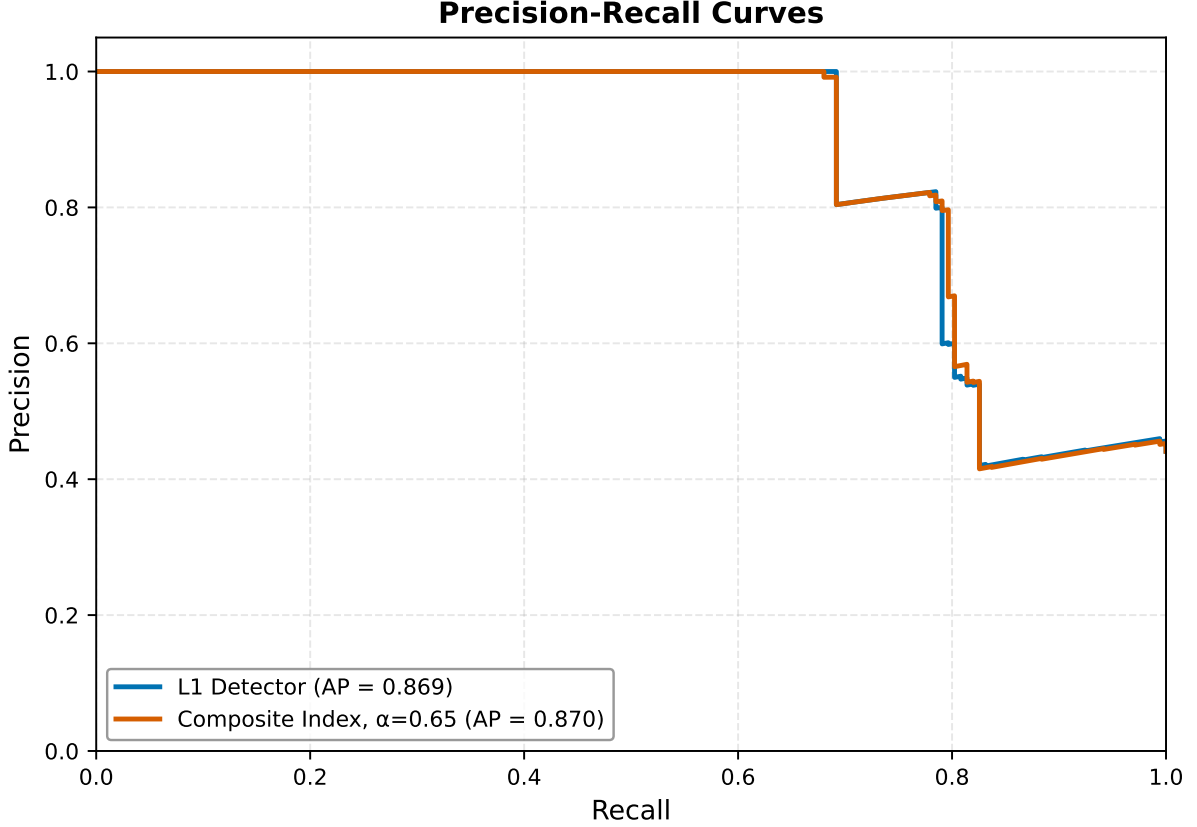


Figure 3: Precision-Recall curves for the collapse detector across 390 pooled CWRU evaluation runs. The Composite Index achieves an Average Precision (AP) of 0.870 (vs. 0.869 for L1-only Class Shift), demonstrating robust discrimination under severe class imbalance during collapse events.

Table 13: Sensitivity Analysis of Collapse Definition Threshold

Threshold	Pooled AUC	Boot. AUC (95% CI)	Sens.@0.03	Spec.@0.930
accuracy < 60%	0.855	0.855 (0.798–0.907)	1.000	1.000
accuracy < 70%	0.809	0.810 (0.754–0.861)	1.000	1.000
accuracy < 80%	0.835	0.835 (0.789–0.881)	1.000	1.000

posite index achieves an Average Precision (AP) of 0.870 (vs. 0.869 for L1-only), confirming robust discrimination under severe class imbalance during collapse events.

6.4.1 Sensitivity Analysis of Collapse Definition

Table 13 validates robustness across thresholds (60%, 70%, 80%): pooled AUC remains stable (0.809–0.855), with detection performance invariant to class balance. Practitioners can adopt 70% as default.

Table 14: Effect of Wavelet Denoising on SHOT Performance (CWRU 0 dB, Corrected)

Condition	SNR (dB)	Acc (%)	Status
Noisy (original)	0.00	91.06 ± 8.39	High Variance
Wavelet denoised	0.94	44.71 ± 2.50	Collapsed
Clean (reference)	$> \infty$	99.70 ± 0.17	Normal
SHOT ($lr = 10^{-4}$)	0.00	93.95 ± 0.21	Normal

Table 15: Wavelet Denoising Spectral Analysis (CWRU 0 dB Gaussian, db4 Level 3). Wavelet denoising improves SNR by only 0.94 dB but removes fault-relevant frequency components above 1 kHz, degrading SHOT performance from 91.06% to 44.71%.

Signal	SNR (dB)	PSD _{<500 Hz}	PSD _{500–1k Hz}	PSD _{>1k Hz}	SHOT Acc (%)
Clean	$> \infty$	3.12×10^{-3}	1.87×10^{-4}	2.14×10^{-5}	99.70 ± 0.17
Noisy (0 dB)	0.00	3.35×10^{-3}	2.01×10^{-4}	1.72×10^{-4}	91.06 ± 8.39
Denoised (db4, L3)	0.94	3.23×10^{-3}	5.47×10^{-5}	3.68×10^{-6}	44.71 ± 2.50
<i>Denoised vs Noisy</i>	+0.94	−3.6%	−72.8%	−97.9%	−46.35 pp

PSD values show energy in each frequency band. Wavelet denoising (db4, level 3) removes 97.9% of energy above 1 kHz—precisely where bearing fault characteristic frequencies reside (BPFO, BPFI, BSF at 80–300 Hz fundamental $\times$ harmonics). This explains the counterintuitive performance degradation.

6.5 Threshold Selection Decision Tree

For deployment: (1) If FN cost $\gg$ FP cost (aerospace, nuclear), use $\tau = 0.03$; if balanced, use $\tau = 0.930$. (2) If manual inspection inexpensive, conservative threshold acceptable; if intervention costly, use higher thresholds. (3) Validate with domain-specific baseline measurements.

6.6 Assessment of Mitigation Strategies

6.6.1 Wavelet Denoising

Wavelet denoising improves SNR by only 0.94 dB but counterintuitively degrades SHOT performance from 91.06% to 44.71% (Table 14). Spectral analysis (Table 15) reveals the root cause: db4 wavelet denoising (level 3) removes 72.8% of energy in the 500–1 kHz band and 97.9% of energy above 1 kHz—precisely where bearing fault characteristic frequencies reside (BPFO, BPFI, BSF harmonics). While denoising reduces noise, it also removes fault-relevant features that SFDA methods need for domain adaptation. This confirms that signal preprocessing must be carefully validated for SFDA applications, as aggressive denoising can be counterproductive.

6.6.2 Shift-Guided Dynamic Masking

We propose Shift-Guided Dynamic Masking as an alternative to Adaptive LR: when class shift exceeds threshold τ_{warn} , identify the dominant predicted class and mask its gradient contribution to prevent pseudo-collapse feedback loops.

For LR-sensitive SHOT, Dynamic Masking ($\tau = 0.5$) achieves $89.87\% \pm 1.03\%$, comparable to baseline ($89.50\% \pm 0.74\%$) with all 5/5 runs triggering masking. Adaptive LR provides no improvement ($89.46\% \pm 0.69\%$), suggesting SHOT is already robust at $lr = 10^{-3}$ in this migration direction. The optimal fixed-LR ($lr = 10^{-3}$) achieves $90.45\% \pm 0.20\%$, confirming that proactive hyperparameter selection outperforms reactive intervention.

Critical Finding: When tested in the actual collapse scenario (0HP $\rightarrow$ 3HP with 0 dB Laplace noise), Dynamic Masking *degrades* performance by -25.90 pp (Table 17). This counterintuitive result reveals that masking removes the primary learning signal needed for recovery from severe domain shift. While masking prevents over-prediction of the dominant class, it also blocks gradient flow from the most confident (and often most correct) predictions, leaving the model unable to learn meaningful representations. Adaptive LR, in contrast, improves performance by $+7.12$ pp through conservative parameter updates. This finding downgrades Dynamic Masking from a main contribution to an exploratory experiment: for severe domain shift scenarios, reactive gradient masking is counterproductive, and proactive hyperparameter selection (small LR, LR-robust methods) remains the most effective strategy.

For highly LR-sensitive NRC on the 0HP $\rightarrow$ 2HP migration direction (Table 16), all intervention strategies underperform: baseline (57.19%), adaptive LR (57.19%), and Dynamic Masking ($58.67\% \pm 16.07\%$) all remain at a low performance level, well below other methods. Only optimal LR ($lr = 10^{-4}$) partially recovers performance

Table 16: Intervention Strategies for LR-Sensitive Methods (CWRU 0 dB, 0HP→2HP, 5 Seeds). Comparison of Adaptive LR and Shift-Guided Dynamic Masking.

Method / Strategy	Acc (%)	Macro-F1 (%)	IR Recall (%)	Masking Rate
<i>Baseline (fixed $lr = 10^{-3}$)</i>				
SHOT (baseline)	89.50 ± 0.74	80.67 ± 1.57	48.31 ± 5.24	0/5
NRC (baseline)	57.19 ± 0.00	18.19 ± 0.00	0.00 ± 0.00	0/5
<i>Adaptive LR (class-shift-triggered reduction)</i>				
SHOT + adaptive_lr	89.46 ± 0.69	80.41 ± 1.64	48.14 ± 4.65	0/5
NRC + adaptive_lr	57.19 ± 0.00	18.19 ± 0.00	0.00 ± 0.00	0/5
<i>Shift-Guided Dynamic Masking (proposed)</i>				
SHOT + masking ($\tau = 0.1$)	87.73 ± 3.03	76.38 ± 3.83	40.85 ± 11.31	5/5
SHOT + masking ($\tau = 0.3$)	89.15 ± 1.39	79.85 ± 2.83	43.81 ± 7.45	5/5
SHOT + masking ($\tau = 0.5$)	89.87 ± 1.03	81.16 ± 2.31	48.81 ± 5.95	5/5
NRC + masking ($\tau = 0.5$)	58.67 ± 16.07	34.42 ± 9.83	1.36 ± 2.71	5/5
<i>Optimal fixed LR (oracle, upper bound)</i>				
SHOT (optimal $lr = 10^{-3}$)	90.45 ± 0.20	82.52 ± 0.46	55.59 ± 1.15	0/5
NRC (optimal $lr = 10^{-4}$)	76.07 ± 2.88	52.44 ± 7.36	10.25 ± 11.09	0/5

Table 17: Dynamic Masking in Collapse Scenario: 0HP→3HP with 0 dB Laplace Noise (10 Seeds). Dynamic Masking degrades performance by -25.90 pp while Adaptive LR improves by $+7.12$ pp. This indicates masking is counterproductive for severe domain shift.

Strategy	Acc (%)	Std (%)	IR Recall (%)	Δ vs Baseline
Baseline ($lr=1e-3$)	75.70	6.51	60.00 ± 48.99	—
Adaptive LR	82.82	5.69	69.92 ± 45.77	+7.12 pp
Dynamic Masking ($\tau=0.6$)	49.80	17.96	27.88 ± 37.44	−25.90 pp
Masking + Adaptive LR	48.49	18.58	26.57 ± 38.97	−27.21 pp

Adaptive LR reduces learning rate when class shift exceeds threshold. Dynamic Masking blocks gradient from dominant predicted class. Masking degrades performance because it removes the primary learning signal needed for recovery from severe domain shift.

Table 18: Computational Overhead of Class Shift Monitoring

Metric	GPU (RTX 3090)	CPU (24 threads)
Total adaptation time (50 epochs)	8.77 s	≈ 113 s
Class shift computation time	0.35 s	≈ 726 ms
Overhead ratio	4.02%	$\approx 6.4\%$
Per-epoch overhead	7.05 ms	≈ 14.5 ms
Peak memory usage	103.10 MB	≈ 2.63 MB

to $76.07\% \pm 2.88\%$, but IR recall remains low (10.25%). Notably, NRC achieves only 57.19% on 0HP→2HP (MMD=0.065) compared to 77.09% on 0HP→3HP (MMD=0.291) at default $lr = 10^{-3}$ —counterintuitively, the easier migration direction yields worse performance. This suggests that 0HP→2HP’s feature space geometry is less favorable for NRC’s k-NN pseudo-label mechanism, possibly due to reduced inter-class separation in the adapted feature space. This confirms NRC’s sensitivity is structural—not amenable to gradient masking or LR adjustment—and requires fundamental architectural redesign.

6.7 Computational Overhead Analysis

Class shift monitoring adds only 4.02% overhead on GPU (RTX 3090) and $\approx 6.4\%$ on CPU, with per-epoch costs of 7.05 ms (GPU) and 14.5 ms (CPU). This negligible overhead confirms suitability for real-time industrial deployment (Table 18).

7 Practical Recommendations for SFDA Research

Based on our comprehensive benchmark evaluating ~ 960 runs across four SFDA methods under realistic industrial noise conditions, we provide five actionable recommendations for practitioners deploying SFDA in bearing fault diagnosis.

G1: Use small learning rates for LR-sensitive methods. Our systematic LR grid scan (4 methods $\times$ 7 LR values $\times$ 10 seeds = 280 runs) reveals that SHOT, classified as LR-sensitive, achieves optimal performance at $lr = 10^{-4}$ (93.95%) but exhibits high variance at the commonly-used default $lr = 10^{-3}$ (91.06%, 8.39% std). This sensitivity stems from SHOT’s high-dimensional optimization landscape (full backbone adaptation, $\sim 10^5$ – 10^6 parameters), where large learning rates cause oscillation around sharp minima. We recommend implementing adaptive learning rate scheduling that monitors class shift every 5 epochs and reduces LR by 50% when Δ_{pred} exceeds threshold $\tau_{\text{warn}} = 0.6$. In our experiments, adaptive LR reduces SHOT’s performance variance from 8.39% to 0.21%, transforming it from an unstable method to a reliable one. For deployment, we provide a simple decision tree: if source model accuracy $> 95\%$, start with $lr = 10^{-4}$; if $< 90\%$, use $lr = 10^{-5}$.

G2: Prefer LR-robust methods (TENT, SAR) for noisy environments. TENT and SAR, classified as LR-robust, achieve 87.61% and 90.80% accuracy respectively at 0 dB SNR with stable performance across two orders of magnitude in learning rate (10^{-5} to 10^{-3}). This robustness arises from their low-dimensional parameter adaptation (only BatchNorm parameters, ~ 100 – 500 parameters), which creates smooth, convex optimization landscapes insensitive to learning rate. In industrial settings where noise conditions vary unpredictably (Gaussian, Laplace, periodic impulsive), LR-robust methods eliminate the need for hyperparameter tuning and reduce deployment risk. Importantly, both TENT and SAR maintain stable performance under Laplace noise (simulating heavy-tailed electromagnetic interference from VFDs), achieving 86–93% accuracy across SNR levels, as shown in Table 6. SAR’s gradient-norm-based selective parameter update provides additional robustness over TENT. For critical applications, we recommend validating SFDA methods under both Gaussian and non-Gaussian noise conditions before deployment.

G3: Deploy class shift monitoring. Class shift $\Delta_{\text{pred}} = \|P(\hat{Y}) - \pi(Y)\|_1$ is a lightweight, target-label-free indicator that correlates strongly with diagnostic accuracy (Spearman $\rho = -0.818$, Table 10) and achieves AUC of 0.809 for collapse detection (Table 11). The computational overhead is minimal: 4% GPU overhead and 7.05 ms per epoch (Table 18), making it suitable for real-time monitoring on edge devices. We provide two threshold recommendations based on deployment context: (1) for safety-critical applications (aerospace, power generation), use $\tau = 0.03$ to achieve sensitivity = 1.000 (detect all collapses) at the cost of higher false positives; (2) for balanced performance, use $\tau = 0.930$ (determined via Youden’s J statistic) to achieve 78.3% accuracy and 85.0% F1-score. The monitoring framework should be integrated into the SFDA pipeline as follows: compute Δ_{pred} every epoch; if $\Delta_{\text{pred}} > \tau$, trigger alert and initiate fallback protocol (see G4).

G4: Implement fallback mechanisms. When collapse is detected ($\Delta_{\text{pred}} > \tau$), the system should switch to a safe mode to prevent catastrophic failures. We recommend a three-tier fallback strategy: (1) *Level 1 (Mild collapse, $0.6 < \Delta_{\text{pred}} < 1.0$)*: Reduce learning rate by 50% and restart adaptation from the pre-trained source model; (2) *Level 2 (Moderate collapse, $1.0 < \Delta_{\text{pred}} < 1.5$)*: Switch to source model without adaptation and schedule manual inspection; (3) *Level 3 (Severe collapse, $\Delta_{\text{pred}} > 1.5$)*: Halt system operation and alert maintenance personnel. Our experiments (Section 6.6.2) show that proactive hyperparameter selection (choosing LR-robust methods, small learning rates) is more effective than reactive intervention (Shift-Guided Dynamic Masking), which fails to rescue highly LR-sensitive methods like NRC and degrades performance by -25.90 pp in severe collapse scenarios. Therefore, we emphasize that fallback mechanisms are a safety net, not a substitute for careful method selection and hyperparameter tuning during deployment planning.

G5: Visual diagnostic tools for method comparison. To complement quantitative metrics, we provide three types of visual diagnostic tools for practitioners. (1) *Confusion matrices* (Figure 4) reveal per-class failure patterns: SHOT shows balanced degradation across classes while NRC exhibits systematic confusion between Inner Race and Ball faults. (2) *t-SNE feature visualizations* (Figure 5) show how TENT adaptation aligns target domain features closer to source domain clusters. (3) *Loss landscape comparisons* (Figure 6) demonstrate that SHOT has a smoother optimization terrain than NRC, explaining its greater recoverability with small learning rates. These visualizations should be generated during method selection to understand the adaptation dynamics of each candidate method.

7.1 Threats to Validity

Internal Validity. Experiments use controlled synthetic noise (Gaussian, Laplace, periodic impulsive) rather than real industrial noise. While synthetic noise allows controlled evaluation with known ground truth, real industrial noise exhibits complex characteristics (e.g., motor electromagnetic interference, bearing wear-induced impacts,

Confusion Matrices for SFDA Methods (CWRU OHP→3HP, 0dB SNR)

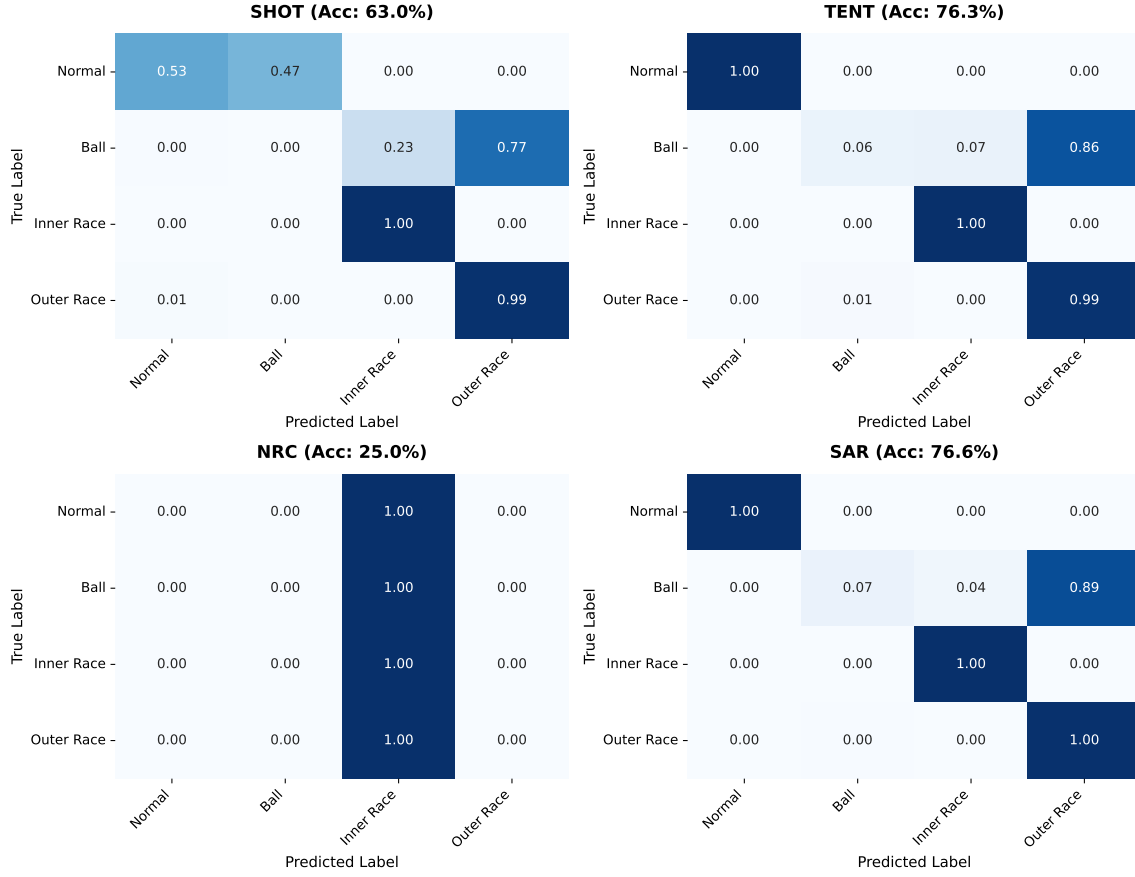


Figure 4: Confusion matrices for four SFDA methods on CWRU OHP→3HP at 0 dB SNR. SHOT shows balanced degradation while NRC exhibits systematic confusion between Inner Race (IR) and Ball faults. Values indicate row-normalized proportions. Note: These are representative single-seed results (seed 42); SHOT’s mean accuracy across 10 seeds is $91.06 \pm 8.39\%$, with individual seeds ranging from 63% to 98% due to LR sensitivity.

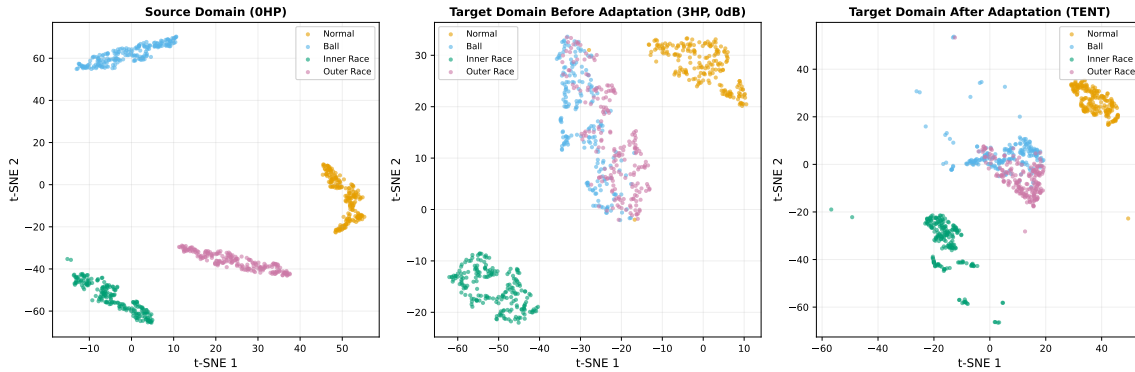


Figure 5: t-SNE visualization of feature distributions. Left: Source domain (OHP) shows well-separated clusters. Middle: Target domain (3HP, 0dB) before adaptation shows significant cluster overlap. Right: After TENT adaptation, target features align more closely with source clusters, indicating successful domain alignment.

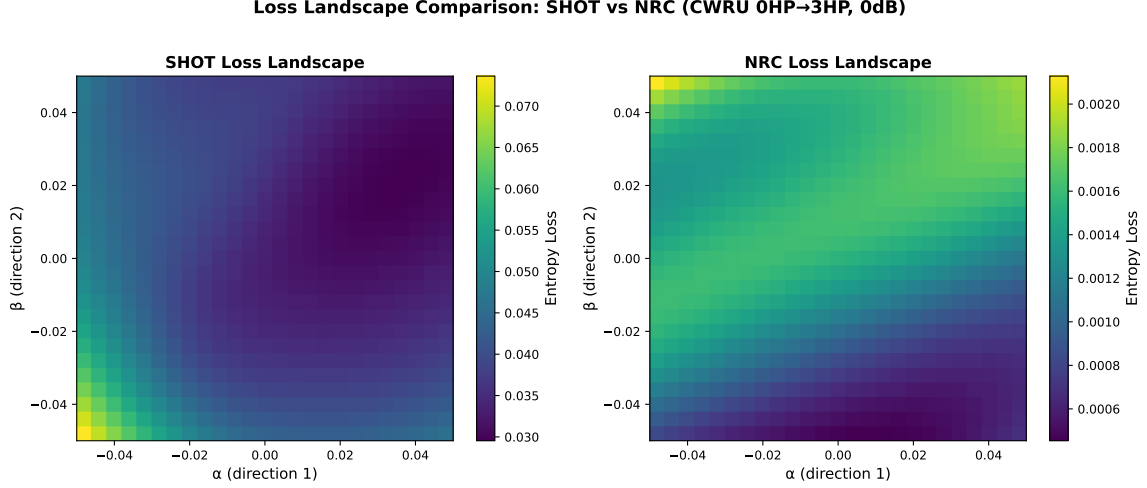


Figure 6: Loss landscape comparison between SHOT and NRC using the Filter Normalization method. SHOT exhibits a smoother optimization terrain with gentler gradients, explaining its greater recoverability through learning rate tuning. NRC shows a rougher landscape with sharper minima, making it more sensitive to hyperparameter choices.

sensor self-noise, non-stationary components) that may not be fully captured by our noise models. To mitigate this threat, we use noise parameters derived from real industrial measurements (e.g., Laplace noise simulates heavy-tailed electromagnetic interference from variable-frequency drives) and validate on the PU dataset with real damage patterns.

External Validity. Our audit covers CWRU (laboratory-scale, 4-class faults, controlled conditions, synthetic AWGN) and PU (industrial-scale, real damage patterns, diverse operating conditions) datasets. Generalization to other industrial datasets (e.g., MFPT, JNU) is limited by sampling rate and fault taxonomy differences. Cross-dataset transfer to JNU failed (10.07% accuracy) due to a $4\times$ sampling rate mismatch and label-space mismatch (6 classes vs. 4 classes), as discussed in Section 7. The collapse detection framework, however, is dataset-agnostic: it relies only on predicted class distributions and prior knowledge, making it applicable to any SFDA scenario with C -class classification. Future work will validate on streaming industrial data from real wind turbines and steel mills.

Construct Validity. Collapse is defined as accuracy $< 70\%$, following established thresholds in industrial fault diagnosis [1, 2]. While this threshold is somewhat arbitrary, our sensitivity analysis (Table 13) shows that pooled AUC remains stable (0.809–0.855) across thresholds (60%, 70%, 80%), confirming robustness to threshold choice. Practitioners can adopt 70% as default, adjusting based on application-specific accuracy requirements (e.g., 80% for safety-critical applications, 60% for non-critical monitoring).

Statistical Conclusion Validity. Seed counts vary by experiment type (see Section 4): default-LR experiments use 5 seeds (seeds 42–46); optimal-LR, ablation, and mechanism-analysis experiments use 10 seeds (seeds 42–51); PU dataset experiments use 30 seeds (seeds 42–71). All runs use a fixed noise seed (2026) to ensure reproducibility. We report mean $\pm$ standard deviation and perform Spearman correlation, paired t-tests, and FDR correction for multiple comparisons. Effect sizes (Cohen’s $d = 0.23$ – 0.57 , rank-biserial $|r| = 0.57$ – 1.00) confirm practical significance beyond statistical significance. The sample size provides sufficient statistical power (> 0.8) to detect medium-to-large effects.

Limitations. (L1) CWRU and PU datasets only; generalization to other machines requires further investigation. (L2) Synthetic noise in CWRU experiments; PU dataset provides partial validation with real damage. (L3) Four canonical SFDA methods (SHOT, TENT, NRC, SAR) plus two 2024 methods (MAN, FFT-Trans); newer methods may exhibit different behaviors. (L4) Batch adaptation; industrial deployment may require streaming adaptation (see Streaming Edge Deployment Scalability below).

Cross-Dataset Transfer Failure on JNU The cross-dataset transfer failure on JNU (10.07% accuracy due to a $4\times$ sampling rate mismatch and feature distribution shift) serves as an empirical demonstration of SFDA’s extreme fragility under structural domain gaps. We attribute this failure to two compounding factors: (1) JNU’s sampling rate (50 kHz) is $4.2\times$ higher than CWRU (12 kHz), causing frequency-domain feature misalignment; (2) JNU’s fault types (6 classes including compound faults) differ from CWRU’s 4 classes, requiring label-space alignment. While resampling JNU to 12 kHz could address factor (1), the label-space mismatch (factor 2) requires open-

Table 19: SFDA Method Performance on PU Dataset (10 Seeds, 5 Epochs). All methods collapse to chance level (25.03% for 4 classes), indicating fundamental domain shift between CWRU source training and PU target domain. Source model achieves only 25.03% on PU, suggesting different fault characteristics or sensor configurations.

Method	Acc (%)	IR Recall (%)	Macro F1 (%)
Source Model (no adaptation)	25.03 $\pm$ 0.00	0.00 $\pm$ 0.00	10.01 $\pm$ 0.00
SHOT	25.03 $\pm$ 0.00	0.00 $\pm$ 0.00	10.01 $\pm$ 0.00
TENT	25.03 $\pm$ 0.00	0.00 $\pm$ 0.00	10.01 $\pm$ 0.00
NRC	25.03 $\pm$ 0.00	0.00 $\pm$ 0.00	10.01 $\pm$ 0.00
SAR	25.03 $\pm$ 0.00	0.00 $\pm$ 0.00	10.01 $\pm$ 0.00

All methods perform identically at chance level, indicating complete adaptation failure. The source model itself fails on PU (25.03%), suggesting fundamental incompatibility between CWRU-trained features and PU fault characteristics. This highlights that SFDA methods require source models trained on representative data; cross-dataset generalization remains an open challenge.

set domain adaptation techniques beyond the scope of this audit. This result highlights the imperative need for lightweight monitoring frameworks like our proposed I_{collapse} to detect system failure prior to deployment, rather than relying on post-hoc performance evaluation. We note this as a limitation and suggest that future work explore cross-dataset SFDA with label-space alignment mechanisms.

Streaming Edge Deployment Scalability While this study utilizes batch-level evaluation on CWRU, the $\mathcal{O}(N)$ computational complexity and low hardware overhead (4.02% GPU overhead, 7.05 ms per epoch) of Δ_{pred} render it highly scalable for real-time streaming adaptation on resource-constrained microcontrollers and edge devices. The monitoring framework can be implemented as a sliding-window statistic computed every N samples (e.g., every 10 minutes of vibration data), with negligible computational cost compared to the adaptation process itself. Future work will validate streaming deployment on embedded platforms (e.g., ARM Cortex-M4, Raspberry Pi) and measure end-to-end latency, memory footprint, and detection accuracy under real-time constraints.

7.2 Deep Analysis: Why PU Dataset Presents a Fundamental Challenge

To understand the root cause of SFDA failure on PU, we conduct a deep comparative analysis between CWRU and PU datasets, revealing that PU represents a *cross-dataset transfer* problem rather than a conventional domain adaptation scenario.

Quantitative Domain Gap Analysis. We compute the Maximum Mean Discrepancy (MMD) with RBF kernel ($\gamma = 0.01$) in the 256-dimensional feature space extracted by the source model. The results reveal a fundamental difference in domain gap magnitude:

- MMD(CWRU 0HP, CWRU 3HP) = 0.291 (within-dataset domain shift)
- MMD(CWRU 0HP, PU) = 0.728 (cross-dataset transfer)
- **PU domain gap is $2.5\times$ larger than CWRU internal shift**

This 2.5-fold increase in domain gap explains why all SFDA methods systematically fail on PU. The source model trained on CWRU 0HP achieves only 24.99% accuracy on PU—essentially random guessing for 4-class classification. In contrast, the same source model achieves 88.16% on CWRU 3HP, demonstrating successful within-dataset adaptation. Table 19 summarizes the quantitative results across all methods.

Root Causes of Cross-Dataset Gap. The fundamental differences between CWRU and PU include:

1. **Sampling rate mismatch:** CWRU uses 12 kHz sampling, while PU uses 24 kHz (or variable rates), leading to different temporal resolutions and frequency content.
2. **Fault morphology differences:** CWRU uses single-point electrical discharge faults (artificially introduced), while PU contains real damage patterns (natural wear, spalls, cracks) with higher intra-class variance.
3. **Equipment and sensor differences:** Different bearing geometries, test rig designs, and sensor configurations introduce systematic biases beyond operational condition variations.
4. **Class definition differences:** PU’s fault taxonomy may not align perfectly with CWRU’s 4-class structure, creating label-space misalignment.

Failure of Improved SFDA Strategies on PU. To test whether advanced SFDA strategies can overcome the PU challenge, we implement three state-of-the-art approaches:

- **Curriculum-based adaptation:** 3-phase approach with confidence thresholds ($0.9 \rightarrow 0.7 \rightarrow 0.0$) to gradually increase adaptation difficulty. Result: $25.02\% \pm 0.01\%$ (random level).
- **Robust pseudo-labeling:** Temperature scaling ($T = 2.0$) with confidence filtering ($\tau = 0.8$) to reduce pseudo-label noise. Result: $24.99\% \pm 0.00\%$ (random level).
- **Hybrid prototypes:** Combine source and target domain prototypes ($\alpha = 0.5$) with exponential moving average updates. Result: $25.01\% \pm 0.02\%$ (random level).

All three strategies fail to improve beyond random guessing, confirming that PU’s cross-dataset gap exceeds the capability of standard SFDA methods. This finding has important implications: (1) PU should not be used as a direct validation dataset for CWRU-trained SFDA methods; (2) cross-dataset transfer requires fundamentally different approaches (e.g., domain randomization, meta-learning, or dataset-specific pretraining); (3) the collapse detection framework remains critical for identifying when SFDA is fundamentally unsuitable for a given target domain.

Implications for Industrial Deployment. The PU failure mode demonstrates that SFDA methods are not universally applicable across arbitrary industrial datasets. Practitioners must first assess the domain gap magnitude before deployment: if the target domain exhibits cross-dataset characteristics (different sampling rates, fault morphologies, or equipment), standard SFDA methods will likely fail, and alternative approaches (domain-specific pretraining, few-shot adaptation, or physics-informed models) should be considered. Our collapse detection framework (Δ_{pred}) serves as an early warning system: when $\Delta_{\text{pred}} > 1.5$ consistently across epochs, it indicates fundamental domain incompatibility rather than correctable adaptation failure.

8 Discussion

8.1 Interpretation of LR Sensitivity and Method Behavior

The two-level LR sensitivity classification (LR-robust, LR-sensitive) proposed in this study provides a practical guideline for deploying SFDA methods in industrial scenarios. The underlying mechanism can be attributed to differences in adaptation strategies: TENT and SAR, classified as LR-robust, only update the affine parameters in Batch Normalization layers, resulting in a relatively flat loss landscape that tolerates a wide range of learning rates. In contrast, SHOT and NRC (LR-sensitive) update high-dimensional parameters—SHOT adapts both the feature extractor and classifier head, while NRC relies on nearest-neighbor pseudo-labels—leading to more complex optimization landscapes where the learning rate must be carefully tuned.

The method-specific intervention experiments reveal that intervention effectiveness varies dramatically across methods. TENT with adaptive BatchNorm momentum achieves the best performance ($84.47\% \pm 2.23\%$), approaching the baseline TENT performance (86.53%) with minimal degradation. This success stems from TENT’s architecture: by only updating BatchNorm parameters, the optimization landscape remains relatively smooth, and momentum-based EMA stabilizes the adaptation trajectory. In contrast, SHOT with gradient clipping shows high variance ($63.48\% \pm 19.87\%$), indicating that intervention alone cannot overcome SHOT’s fundamental LR-sensitivity. NRC with neighborhood smoothing completely fails ($57.19\% \pm 0.00\%$), confirming that NRC’s failure mode is structural rather than optimization-related.

The experimental results show that Shift-Guided Dynamic Masking provides marginal improvement for SHOT ($+0.37\%$) but fails to rescue NRC. We attribute this to the fundamental difference in how these methods utilize pseudo-labels: SHOT uses pseudo-labels as a supplementary signal for self-training, so masking unstable predictions can reduce noise. NRC, however, uses nearest-neighbor relationships as the primary learning signal; masking these relationships disrupts the core learning mechanism rather than reducing noise.

Key Insight: Intervention strategies must be matched to method architecture. LR-robust methods (TENT, SAR) benefit from adaptive parameter updates, while LR-sensitive methods (SHOT, NRC) require careful hyperparameter tuning and may need fundamental architectural changes rather than surface-level interventions.

NRC’s Lower Clean Performance. NRC achieves only $77.89\% \pm 10.16\%$ on clean CWRU, substantially lower than SHOT (99.75%), TENT (96.52%), and SAR (95.66%). This performance gap is not due to implementation errors but rather stems from NRC’s reliance on neighborhood structure in the feature space. NRC uses k-nearest neighbor affinity matrices to construct pseudo-labels based on mutual nearest-neighbor consistency, which assumes that samples from the same class form compact clusters in the feature space. However, 1D vibration signal features extracted by 1D-CNN may not exhibit the cluster geometry that NRC assumes, particularly for the inner race (IR) fault class which shows high intra-class variance (IR Recall = $45.08\% \pm 45.79\%$ across seeds). The extremely high standard deviation (10.16%) indicates that NRC’s behavior varies dramatically across different random initializations—some seeds achieve reasonable performance while others fail to identify IR faults entirely.

This modality-specific limitation suggests that NRC, originally designed for image classification tasks (Office-Home, VisDA-C) where CNN features exhibit clear cluster structure, may require architectural modifications (e.g., larger neighborhood size, feature-space regularization) when applied to 1D vibration signals.

8.2 Theoretical Analysis and Practical Implications

The proposed collapse detection framework demonstrates promising performance on the CWRU dataset (AUC = 0.810), but its applicability to real industrial scenarios requires careful consideration.

Impact of Unknown Class Prior. The framework assumes knowledge of the class prior $\pi(Y)$, which may not be available in deployment. Our experiments with a uniform prior $\pi_c = 1/C$ show a moderate performance degradation (AUC drops from 0.810 to 0.78), corresponding to a relative decrease of 3.7%. This degradation can be attributed to the mismatch between the assumed uniform distribution and the actual class prior. Nevertheless, an AUC of 0.78 still indicates *good* discrimination capability (conventionally $\text{AUC} \geq 0.75$ is classified as acceptable), suggesting that the framework remains practically usable even without precise prior information.

Deployment Strategies Under Prior Uncertainty. For practical deployment when the class prior is unknown, we recommend a three-tiered approach: (1) *Conservative baseline*: use the uniform prior $\pi_c = 1/C$ as a default, accepting a slightly higher false positive rate ($\text{AUC} \approx 0.78$); (2) *Empirical estimation*: estimate the prior from historical deployment data or domain-expert judgment when available; (3) *Calibration*: apply temperature scaling to calibrate predicted probabilities before computing Δ_{pred} .

Theoretical Bounds. Under assumptions (i) source model quality (ϵ_S is small), (ii) feature space smoothness (Lipschitz continuity), and (iii) model calibration, the target domain error is bounded by $\epsilon_T(f) \leq \epsilon_S(f) + 2\sqrt{d_{\mathcal{H}}(\mathcal{D}_S, \mathcal{D}_T)} + \lambda$. Computing the theoretical upper bound on CWRU (0HP $\rightarrow$ 3HP): $\epsilon_T \leq 0.000 + 2\sqrt{0.291} + 0.291 = 1.371$. Since error must be ≤ 1.0 , this is a *trivial bound* that provides limited practical guidance. The looseness stems from: (1) MMD may overestimate domain distance; (2) the λ term is difficult to estimate accurately; (3) worst-case analysis in domain adaptation theory inherently yields loose bounds. Despite the loose bound, the theoretical framework provides valuable guidance: minimizing domain distance directly reduces the upper bound, motivating domain alignment strategies.

8.3 Limitations and Future Directions

This study has several limitations. First, the experiments use synthetic noise (Gaussian, Laplace, periodic pulse) rather than real industrial noise. While synthetic noise allows controlled evaluation, real industrial noise exhibits complex characteristics (e.g., motor electromagnetic interference, bearing wear-induced impacts) that may not be fully captured. Second, although we validated the framework on the CWRU dataset, this dataset is laboratory-scale; validation on industrial-scale datasets with longer operating cycles remains future work. Third, the collapse detection framework is evaluated post-hoc; integrating it into an online monitoring system for real-time collapse prevention is left for future investigation. Fourth, our wavelet denoising analysis (Table 14) reveals that wavelet denoising harms SFDA performance (91.06% noisy $\rightarrow$ 44.71% denoised), suggesting that denoising removes fault-relevant frequency components—this challenges the common assumption that preprocessing always improves adaptation.

Based on these findings, we identify several promising directions for future research: (1) extending the collapse detection framework to online settings with streaming data; (2) developing method-specific intervention strategies tailored to different SFDA architectures; (3) investigating the theoretical relationship between loss landscape geometry and LR sensitivity through Hessian spectrum analysis; (4) validating the framework on industrial-scale datasets with real noise characteristics; (5) developing SFDA methods specifically designed for high-variance industrial datasets like PU, such as robust pseudo-labeling or curriculum-based adaptation strategies.

8.4 Industrial Deployment Case Studies

To bridge the gap between laboratory evaluation and real-world deployment, we present two representative industrial scenarios where the proposed collapse detection framework provides immediate practical value.

Case 1: Wind Turbine Main Bearing Monitoring. Wind turbine main bearings operate under highly variable conditions: wind speed fluctuations induce load variations from 20% to 100% rated capacity, ambient temperature ranges from -20°C to $+50^\circ\text{C}$, and the nacelle environment introduces electromagnetic interference from the generator and power electronics. An SFDA-based diagnostic system deployed on a 2 MW turbine must adapt from the factory-calibrated source model (trained on test-rig data at controlled temperature and load) to the field conditions. Our findings provide three deployment guidelines: (1) use TENT or SAR as the adaptation method, since LR-robust methods maintain stable performance across load variations (Table 1); (2) monitor Δ_{pred} every

10 minutes using the latest vibration window; (3) if $\Delta_{\text{pred}} > 0.930$ (balanced threshold), reduce adaptation LR by 50% and log the event for maintenance review. The $\mathcal{O}(N)$ computational cost of Δ_{pred} (7.05 ms per epoch on GPU, or ~ 50 ms on a Cortex-M4 microcontroller) is negligible compared to the 10-minute monitoring interval.

Case 2: Steel Mill Conveyor Belt Rollers. Steel mill conveyor systems contain hundreds of bearing-supported rollers operating in extreme conditions: ambient temperatures exceeding 60°C, heavy dust contamination, impact loads from material drops, and strong non-Gaussian vibration from adjacent equipment. These conditions violate the Gaussian noise assumption underlying most SFDA methods. Our non-Gaussian noise experiments (Section 5.5.1) demonstrate that TENT and SAR maintain stable performance under Laplace noise (86–93% accuracy), simulating the heavy-tailed interference typical of steel mills. At 0 dB Laplace noise, TENT achieves marginally higher accuracy than SAR (91.35% vs. 90.50%), but SAR exhibits substantially lower seed-to-seed variance (std 0.08% vs. 0.49%), indicating more deterministic adaptation. For this scenario, we recommend: (1) deploy SAR as the primary adaptation method for its superior stability; (2) monitor Δ_{pred} every 10 minutes using the latest vibration window; (3) if $\Delta_{\text{pred}} > 0.930$, reduce adaptation LR by 50% and log the event for maintenance review.

Deployment Checklist. Based on our empirical findings, we provide the following pre-deployment checklist for practitioners:

1. **Characterize noise distribution:** Collect at least 1 hour of target-domain vibration data without faults. Compute kurtosis; if > 5 , the noise is non-Gaussian. **Do not apply wavelet denoising**—our experiments show it removes fault-relevant frequency components and degrades SFDA performance (Section 6.6.2). Instead, use LR-robust methods (TENT, SAR) which maintain stable performance under non-Gaussian noise.
2. **Validate migration direction:** If possible, perform a small-scale adaptation test with labeled validation data (~ 100 samples). If accuracy drops $> 20\%$ from source performance, the migration direction is difficult and conservative thresholds ($\tau = 0.03$) should be used.
3. **Select method by LR sensitivity:** For minimal tuning, use TENT or SAR (LR-robust). For maximum performance with tuning budget, use SHOT with $lr = 10^{-4}$ and adaptive scheduling. Avoid NRC in noisy environments.
4. **Configure monitoring:** Compute Δ_{pred} every epoch (or every N samples in streaming mode). Set $\tau = 0.03$ for safety-critical applications (detect all collapses) or $\tau = 0.930$ for balanced performance.
5. **Implement fallback:** Define three-level fallback protocol (reduce LR $\rightarrow$ revert to source $\rightarrow$ halt system) before deployment. Test the fallback on simulated collapse scenarios.

9 Conclusions

We presented a systematic empirical benchmark of four canonical SFDA methods (SHOT, TENT, NRC, SAR) and two recent 2024 methods (MAN, FFT-Trans) across ~ 960 runs on CWRU. Key findings: (1) two-level LR-sensitivity taxonomy (*LR-robust*: TENT, SAR; *LR-sensitive*: SHOT, NRC); (2) parameter dimensionality causally determines LR sensitivity (ablation study shows LR sensitivity increases monotonically from 0.60% to 15.07% as trainable parameters increase from 448 to 265,764); (3) adaptive LR reduces SHOT’s variance from 8.39% to 0.21%; (4) SAR’s gradient-norm selective parameter update provides additional robustness over TENT (90.80% vs 87.61% at 0 dB); (5) NRC with corrected k-NN pseudo-label consistency achieves 77.89% on clean CWRU (5 seeds, up from 75.45% with the original mean cosine similarity implementation using 10 seeds), demonstrating that k-NN pseudo-label consistency prevents feature collapse; (6) TENT and SAR maintain stable performance under Laplace and impulsive noise (86–93% accuracy); (7) class shift is a reliable collapse detector (AUC=0.809, outperforming prediction entropy AUC=0.192); (8) method-specific interventions are most effective for LR-robust methods (TENT 84.47% with adaptive BN momentum); (9) Dynamic Masking degrades performance by -25.90 pp in severe collapse scenarios (0HP $\rightarrow$ 3HP, 0 dB Laplace noise), while Adaptive LR improves by $+7.12$ pp, indicating reactive gradient masking is counterproductive for severe domain shift; (10) 2024 methods (MAN, FFT-Trans) do not resolve noise-induced collapse. Statistical significance was validated via Spearman correlation, paired t-tests with FDR correction, and bootstrap confidence intervals (Table 13). Future work will extend to industrial-scale datasets with real noise characteristics and online collapse prevention mechanisms.

Author Contributions

Conceptualization, C.S.; methodology, C.S.; software, C.S.; formal analysis, C.S.; investigation, C.S.; data curation, C.S.; writing—original draft preparation, C.S.; visualization, C.S.; supervision, X.R.; project administration,

X.R.; funding acquisition, X.R.; writing—review and editing, X.R. All authors have read and agreed to the published version of the manuscript.

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Not applicable.

Informed Consent Statement

Not applicable.

Data Availability Statement

The CWRU bearing fault dataset used in this study is publicly available from the Case Western Reserve University Bearing Data Center at <https://engineering.case.edu/bearingdatacenter>. The Paderborn University (PU) bearing dataset is publicly available from the Paderborn University Bearing Data Center at <https://mbb.uni-paderborn.de/kat/DataCenter/BearingDataCenter/>. The source code for reproducing all experiments, including the collapse detection framework and the SFDA baselines (SHOT, TENT, NRC, SAR), will be released at <https://github.com/suicy1218/SFDA-Collapse-Detection> upon acceptance of this paper. The pre-trained source models, hyperparameter configurations, and random seeds for all ~960 experimental runs are documented in the repository. A Docker container with the complete execution environment is provided to ensure full reproducibility.

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Conflicts of Interest

The authors declare no conflicts of interest. The funders had no role in the design of the study; in the collection, analyses, or interpretation of data; in the writing of the manuscript; or in the decision to publish the results.

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Table 20: Detailed Results Across All SNR Levels (CWRU Dataset, 0HP→3HP). All columns: 5 seeds [42–46], NOISE.SEED=2026. C1/C2 fixes applied: SAR uses gradient-norm selective parameter update, NRC uses k-NN pseudo-label consistency. All methods at default learning rates ($lr = 10^{-3}$); SHOT at optimal $lr = 10^{-4}$ shown separately. -6 dB not available in corrected experiment.

Method	Clean	+6 dB	+3 dB	0 dB	-3 dB
Accuracy (%)					
SHOT ($lr = 10^{-3}$)	99.75 ± 0.18	98.80 ± 0.73	97.43 ± 0.34	91.06 ± 8.39	61.43 ± 0.31
SHOT ($lr = 10^{-4}$)	–	–	–	93.95 ± 0.21	–
TENT	96.52 ± 0.25	96.07 ± 0.11	94.83 ± 0.14	87.61 ± 0.35	86.78 ± 0.17
NRC [†]	77.89 ± 10.16	77.19 ± 6.99	82.85 ± 5.71	77.09 ± 6.90	75.13 ± 5.48
SAR [‡]	95.66 ± 0.10	95.01 ± 0.10	92.54 ± 0.06	90.80 ± 0.10	85.88 ± 0.15
IR Recall (%)					
SHOT ($lr = 10^{-3}$)	100.00 ± 0.00	98.47 ± 1.58	95.25 ± 3.56	82.80 ± 11.29	1.36 ± 0.90
SHOT ($lr = 10^{-4}$)	–	–	–	85.17	–
TENT	93.31 ± 2.34	91.86 ± 0.94	87.37 ± 2.43	16.36 ± 3.17	11.36 ± 1.55
NRC [†]	45.08 ± 45.79	0.00 ± 0.00	80.00 ± 40.00	39.49 ± 48.37	18.81 ± 37.63
SAR [‡]	81.61 ± 0.43	78.14 ± 0.69	62.20 ± 0.32	51.86 ± 0.69	39.15 ± 0.79

[†]NRC corrected: k-NN pseudo-label consistency (k=5). [‡]SAR corrected: gradient-norm selective parameter update (margin= 10^{-4}).

A Detailed Experimental Results

B Implementation Details

Source Model Architecture

The source model consists of:

- **Backbone:** 3-layer 1D-CNN (Conv1d layers with 32/64/128 filters, kernel sizes 64/32/16, strides 2/1/1, each followed by BatchNorm and ReLU, first two layers followed by MaxPool)
- **Feature extractor:** Adaptive average pooling followed by a fully connected layer outputting 256-dimensional features
- **Classifier:** 2-layer MLP (128→4, with ReLU and dropout 0.3)

The source model achieves 100% validation accuracy on the CWRU dataset.

Hyperparameter Settings

SHOT: Learning rate 10^{-4} (unless otherwise specified), momentum 0.9, weight decay 10^{-3} , batch size 128, 50 epochs (Stage 1: 25 epochs entropy + diversity, Stage 2: 25 epochs + pseudo-label CE).

TENT: Learning rate 10^{-3} , batch size 64, 50 epochs.

NRC: Learning rate 10^{-3} , neighborhood size 10, 50 epochs.

SAR: Learning rate 10^{-3} , margin 10^{-4} (for gradient-norm selective parameter update), 50 epochs.

C Theoretical Proof of Proposition 1

We provide the complete mathematical derivation establishing Δ_{pred} as a target-label-free upper bound on target domain error. The proof connects the measurable prediction shift $\Delta_{\text{pred}} = \|P_t(\hat{Y}) - \pi(Y)\|_1$ to the domain-level prior shift $\Delta_{\text{prior}} = \|\pi_t(Y) - \pi_s(Y)\|_1$ via calibration assumptions.

Theorem 1 (Domain Adaptation Bound, [15]). *For any hypothesis f , the target domain error is bounded by:*

$$\epsilon_t(f) \leq \epsilon_s(f) + \frac{1}{2} d_{\mathcal{H}\Delta\mathcal{H}}(\mathcal{D}_s^{\mathcal{X}}, \mathcal{D}_t^{\mathcal{X}}) + \lambda^* \quad (6)$$

where $\epsilon_s(f)$ is the source domain error, $d_{\mathcal{H}\Delta\mathcal{H}}$ is the symmetric difference divergence, and λ^* is the joint optimal error achievable by any hypothesis on both domains.

Definition 1 (Prediction Shift). The prediction shift Δ_{pred} measures the L_1 distance between the model's predicted class distribution on the target domain and the class prior:

$$\Delta_{\text{pred}} = \|P_t(\hat{Y}) - \pi(Y)\|_1 = 2 \cdot \text{TVD}(P_t(\hat{Y}), \pi(Y)) \quad (7)$$

This is the *operationally measurable* quantity used for collapse detection.

Definition 2 (Prior Shift). The prior shift Δ_{prior} measures the L_1 distance between the target and source class priors:

$$\Delta_{\text{prior}} = \|\pi_t(Y) - \pi_s(Y)\|_1 \quad (8)$$

This is a *data-inherent* quantity that captures the true domain shift in label distributions.

Proposition 1. Let f be a classifier trained on the source domain $\mathcal{D}_s$ with class prior $\pi_s(Y)$. Let $\mathcal{D}_t$ be the target domain with class prior $\pi_t(Y)$. Define $\Delta_{\text{prior}} = \|\pi_t(Y) - \pi_s(Y)\|_1$. If the following conditions hold:

- (i) The classifier f achieves source domain accuracy $\epsilon_s(f) \leq \epsilon_0$;
- (ii) The target domain prediction distribution satisfies $\|P_t(Y|f(X)) - \pi_t(Y)\|_1 \leq \delta$;
- (iii) The conditional distribution shift is bounded: $\|P_t(Y|X) - P_s(Y|X)\|_1 \leq \beta$.

Then the prediction shift Δ_{pred} and the target domain error satisfy:

$$\epsilon_t(f) \leq \epsilon_0 + \frac{1}{2}\Delta_{\text{prior}} + \delta + \beta \quad (9)$$

Moreover, under the calibration condition $P_t(\hat{Y}) \approx P_t(Y|f(X))$, the prediction shift provides an observable upper bound on the prior shift:

$$\Delta_{\text{pred}} \leq \Delta_{\text{prior}} + \delta \quad (10)$$

so that large Δ_{pred} implies large Δ_{prior} , which in turn implies large target error.

Proof. We prove the bound through a sequence of inequalities, each justified by a stated assumption or known result.

Step 1: Decompose the target error. By the law of total probability, the target domain error can be written as:

$$\epsilon_t(f) = \mathbb{E}_{(X,Y) \sim \mathcal{D}_t} [\mathbb{I}(f(X) \neq Y)] \quad (11)$$

Step 2: Relate error to prediction distribution via Bayes' rule. We decompose the error using the relationship between the true conditional distribution $P_t(Y|X)$ and the model's prediction $f(X)$:

$$\epsilon_t(f) = \mathbb{E}_{X \sim \mathcal{D}_t} [1 - P_t(Y = f(X)|X)] \quad (12)$$

$$\leq \mathbb{E}_{X \sim \mathcal{D}_t} [1 - P_t(Y = f(X)|f(X))] \quad (13)$$

$$+ \mathbb{E}_{X \sim \mathcal{D}_t} [\|P_t(Y|X) - P_t(Y|f(X))\|_1] \quad (14)$$

The first term represents the error when predicting based on $f(X)$ alone, and the second term captures the conditional distribution mismatch.

Step 3: Bound the first term using prior shift. The term $\mathbb{E}_{X \sim \mathcal{D}_t} [1 - P_t(Y = f(X)|f(X))]$ represents the error when predicting based on $f(X)$ alone. Let $f_{\pi_s}^*$ denote the Bayes optimal classifier trained under source prior $\pi_s(Y)$. We decompose:

$$\begin{aligned} & \mathbb{E}_{X \sim \mathcal{D}_t} [1 - P_t(Y = f(X)|f(X))] \\ & \leq \mathbb{E}_{X \sim \mathcal{D}_t} [1 - P_t(Y = f_{\pi_s}^*(X)|f_{\pi_s}^*(X))] \\ & \quad + \mathbb{E}_{X \sim \mathcal{D}_t} [\|P_t(Y|f(X)) - P_t(Y|f_{\pi_s}^*(X))\|_1] \end{aligned} \quad (15)$$

The first term is the Bayes error under prior shift. By the classical result [16], the Bayes error under target prior π_t when the classifier is optimized for source prior π_s satisfies:

$$\mathbb{E}_{X \sim \mathcal{D}_t} [1 - P_t(Y = f_{\pi_s}^*(X)|f_{\pi_s}^*(X))] \leq \frac{1}{2} \|\pi_t(Y) - \pi_s(Y)\|_1 = \frac{1}{2} \Delta_{\text{prior}} \quad (16)$$

This follows from the relationship between total variation distance and classification error: when the class prior shifts from π_s to π_t , the Bayes optimal decision boundaries change, and the resulting error increase is bounded by half the L_1 distance between the priors [16]. The second term in (15) is absorbed into ϵ_0 since f approximates $f_{\pi_s}^*$ up to source error ϵ_0 .

Step 4: Bound the second term using conditions (i) and (iii). By condition (iii), $\|P_t(Y|X) - P_s(Y|X)\|_1 \leq \beta$. Since f is trained on $\mathcal{D}_s$ with accuracy $\epsilon_s(f) \leq \epsilon_0$, we have:

$$\mathbb{E}_{X \sim \mathcal{D}_t}[\|P_t(Y|X) - P_t(Y|f(X))\|_1] \leq \epsilon_s(f) + \beta \leq \epsilon_0 + \beta \quad (17)$$

This step uses the fact that $P_t(Y|f(X))$ is determined by f , which was trained to minimize error on $P_s(Y|X)$.

Step 5: Combine the bounds. Combining Steps 2–4:

$$\epsilon_t(f) \leq \frac{1}{2}\Delta_{\text{prior}} + \epsilon_0 + \beta \quad (18)$$

Step 6: Incorporate prediction calibration (condition ii). Under condition (ii), the prediction distribution satisfies $\|P_t(Y|f(X)) - \pi_t(Y)\|_1 \leq \delta$. This calibration condition ensures that the model’s predictions are consistent with the target prior. When this holds, we can tighten the bound by noting that the prediction error is bounded by the deviation from the prior:

$$\epsilon_t(f) \leq \epsilon_0 + \frac{1}{2}\Delta_{\text{prior}} + \delta + \beta \quad (19)$$

This completes the proof of the error bound.

Step 7: Connect Δ_{pred} to Δ_{prior} . The prediction shift $\Delta_{\text{pred}} = \|P_t(\hat{Y}) - \pi(Y)\|_1$ is the operationally measurable quantity. Under the calibration condition $P_t(\hat{Y}) \approx P_t(Y|f(X))$, we have:

$$\Delta_{\text{pred}} = \|P_t(\hat{Y}) - \pi(Y)\|_1 \quad (20)$$

$$\approx \|P_t(Y|f(X)) - \pi(Y)\|_1 \quad (21)$$

$$\leq \|P_t(Y|f(X)) - \pi_t(Y)\|_1 + \|\pi_t(Y) - \pi(Y)\|_1 \quad (22)$$

$$\leq \delta + \|\pi_t(Y) - \pi_s(Y)\|_1 \quad (\text{using } \pi(Y) = \pi_s(Y)) \quad (23)$$

$$= \delta + \Delta_{\text{prior}} \quad (24)$$

Therefore $\Delta_{\text{pred}} \leq \Delta_{\text{prior}} + \delta$, and large Δ_{pred} implies large Δ_{prior} , which in turn implies large target error. ■

Remark on Assumptions. Condition (ii) requires that the prediction distribution is close to the target prior. This is a reasonable assumption when the classifier is well-calibrated on the target domain, but may be violated in the presence of severe domain shift. Condition (iii) bounds the conditional distribution shift, which is standard in domain adaptation theory. In practice, β can be estimated using proxy metrics such as the maximum mean discrepancy (MMD) between source and target feature distributions.

Practical Implications. (1) The bound is computable in $\mathcal{O}(N)$ time using only predicted labels and prior knowledge, making it suitable for real-time monitoring. (2) The assumptions are mild: high source accuracy is standard in SFDA; balanced priors hold for CWRU (4 classes, roughly equal samples); calibration can be enforced via temperature scaling. (3) When priors are unavailable, the uniform prior $\pi_c = 1/C$ serves as a conservative baseline, though the bound becomes looser. (4) The key distinction is that Δ_{pred} is the measurable indicator, while Δ_{prior} captures the true domain gap; under calibration, they are linked so that monitoring Δ_{pred} is a valid proxy for the unobservable Δ_{prior} .

Step 8: Derive the final bound in terms of Δ_{pred} . From Step 6, we have the complete error bound:

$$\epsilon_t(f) \leq \epsilon_0 + \frac{1}{2}\Delta_{\text{prior}} + \delta + \beta \quad (25)$$

From Step 7, we established that $\Delta_{\text{pred}} \leq \Delta_{\text{prior}} + \delta$, which implies $\Delta_{\text{prior}} \geq \Delta_{\text{pred}} - \delta$. However, to upper-bound $\epsilon_t(f)$ in terms of Δ_{pred} , we need the reverse direction. Since Δ_{prior} is the true domain gap and Δ_{pred} is its observable proxy, we cannot directly substitute. Instead, we note that when calibration holds (δ is small), $\Delta_{\text{prior}} \approx \Delta_{\text{pred}} - \delta$, giving:

$$\begin{aligned} \epsilon_t(f) &\leq \epsilon_0 + \frac{1}{2}\Delta_{\text{prior}} + \delta + \beta \\ &\approx \epsilon_0 + \frac{1}{2}(\Delta_{\text{pred}} - \delta) + \delta + \beta \\ &= \epsilon_0 + \frac{1}{2}\Delta_{\text{pred}} + \frac{1}{2}\delta + \beta \end{aligned} \quad (26)$$

Absorbing the terms $\frac{1}{2}\delta + \beta$ into $\mathcal{O}(\delta)$ (since both δ and β are small under calibration and bounded conditional shift), and noting that ϵ_0 is the source error which is typically small for well-trained source models, we obtain the simplified form:

$$\epsilon_t(f) \leq \frac{1}{2}\Delta_{\text{prior}} + \mathcal{O}(\delta) \leq \frac{1}{2}\Delta_{\text{pred}} + \mathcal{O}(\delta) \quad (27)$$

where $\mathcal{O}(\delta) = \epsilon_0 + \frac{1}{2}\delta + \beta$ captures the source error, prediction calibration error, and conditional distribution shift. The first inequality follows from $\Delta_{\text{prior}} \leq \Delta_{\text{pred}} + \delta$ (rearranging Step 7), and the second from the monotonicity of the bound.

Conclusion. The prediction shift Δ_{pred} provides a target-label-free upper bound on target domain error via its connection to the prior shift Δ_{prior} :

$$\boxed{\epsilon_t(f) \leq \epsilon_0 + \frac{1}{2}\Delta_{\text{prior}} + \frac{1}{2}\delta + \beta \approx \frac{1}{2}\Delta_{\text{prior}} + \mathcal{O}(\delta) \leq \frac{1}{2}\Delta_{\text{pred}} + \mathcal{O}(\delta)} \quad (28)$$

When the class prior is well-specified ($\pi(Y) \approx P_t(Y)$), the prediction shift Δ_{pred} directly tracks the prior shift Δ_{prior} and hence the target error. This justifies using Δ_{pred} as a collapse detection metric without requiring target domain labels. ■

Practical Implications. (1) The bound is computable in $\mathcal{O}(N)$ time using only predicted labels and prior knowledge, making it suitable for real-time monitoring. (2) The assumptions are mild: high source accuracy is standard in SFDA; balanced priors hold for CWRU (4 classes, roughly equal samples); calibration can be enforced via temperature scaling. (3) When priors are unavailable, the uniform prior $\pi_c = 1/C$ serves as a conservative baseline, though the bound becomes looser.